

An Explainable Deep Learning Mixture-of-Experts Framework with GAN-Based Data Augmentation for Four-Class Classification of Thyroid Disease Using [^{99m}Tc]Tc-Pertechnetate Scintigraphy: A Multicenter Study

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ABSTRACT

Purpose: Thyroid scintigraphy provides valuable functional information for evaluating thyroid disorders; however, interpretation remains dependent on expert assessment and is challenged by heterogeneous disease patterns and overlapping scintigraphic appearances. Although deep learning (DL) models have shown potential for automated thyroid image analysis, conventional multiclass classifiers rely on shared representations and provide limited interpretability. This study aimed to develop an explainable class-specific Mixture-of-Experts (MoE) framework for multiclass thyroid scintigraphy classification and investigate expert specialization using explainable artificial intelligence (XAI) in a multicenter setting.

Methods: A retrospective dataset of [^{99m}Tc]Tc-pertechnetate thyroid scintigraphy images collected from five nuclear medicine centers was analyzed. Images were categorized into four diagnostic groups: diffuse goiter (n=691), tumoral disease (n=922), thyroiditis (n=423), and normal scintigraphy (n=337). A preprocessing

pipeline including denoising, deblurring, contrast enhancement, resizing, normalization, and training-only augmentation was applied. Conditional GAN-based augmentation was used to address class imbalance, resulting in a final dataset of 3,526 images. Five CNN architectures were evaluated as baseline models. The proposed MoE framework consisted of four class-specific expert networks trained using a one-versus-rest strategy, followed by a trainable fusion network. Model interpretation was performed using Gradient-weighted Class Activation Mapping (Grad-CAM) and expert-response analysis. A binary tumoral versus non-tumoral classification experiment was additionally performed to evaluate the effect of reducing diagnostic complexity.

Results: The MoE framework demonstrated comparable multiclass performance to baseline CNN models, with the VGG11-based MoE achieving the highest fusion accuracy (0.676 ± 0.021). The corresponding expert accuracies were 0.786 ± 0.026 , 0.786 ± 0.016 , 0.950 ± 0.013 , and 0.866 ± 0.012 for diffuse goiter, tumoral disease, thyroiditis, and normal scintigraphy experts, respectively. Grad-CAM visualization demonstrated that predictions were mainly associated with clinically relevant thyroid uptake regions. Binary classification further improved performance, highlighting the impact of diagnostic granularity on classification difficulty.

Conclusion: This study introduces an explainable class-specific MoE framework for multiclass thyroid scintigraphy analysis in a heterogeneous multicenter setting. By enabling specialized learning pathways for different diagnostic categories and providing interpretable expert-level decision analysis, the proposed approach addresses important limitations of conventional CNN-based classification. The findings demonstrate the potential of MoE and XAI strategies to improve transparency and interpretability of AI-assisted nuclear medicine image interpretation. Further external validation on independent multicenter cohorts is required before clinical translation.

Keywords: [^{99m}Tc]Tc-pertechnetate thyroid scintigraphy; convolutional neural networks (CNN); Deep learning (DL); automated classification; Multicenter study; Explainable artificial intelligence (XAI); Mixture of Experts (MoE).

1. INTRODUCTION

The thyroid gland is an important physiological component responsible for producing thyroid hormones, which control the process of metabolism, thermogenesis, energy expenditure, and many other biological functions. Dysfunction in the physiological role of the thyroid gland can adversely affect its ability to regulate these functions and cause various clinical problems, such as hypothyroidism, hyperthyroidism, thyroiditis, and nodular thyroid disease. Disorders of the thyroid gland have become one of the most common endocrine diseases globally, involving many people [1-3].

The proper diagnostic approach to thyroid diseases involves the use of various clinical and imaging techniques, such as physical examination, biochemical measurements of Thyroid-stimulating hormone (TSH) and thyroid hormones, ultrasonography, computed tomography (CT), and nuclear medicine imaging. Of the aforementioned techniques, thyroid scintigraphy offers specific functional characteristics through the assessment of radionuclide uptake and distribution in thyroid tissues. The development of [^{99m}Tc]Tc-pertechnetate thyroid scintigraphy represented an effective solution for the assessment of thyroid trapping function owing to its advantageous physical properties and imaging features [4].

Nevertheless, despite the significance of thyroid scintigraphy in clinical diagnosis, the interpretation process is rather complicated. It is primarily performed visually by a trained nuclear medicine physician, who analyzes the tracer uptake pattern and the gland's structure. Subjective evaluation, different levels of clinical experience, and differences in identification of subtle imaging features may cause inconsistencies in the diagnosis process. Moreover, some thyroid diseases can manifest similar scintigraphy findings, which complicates their reliable differentiation in practice. These limitations have motivated the development of computer-aided diagnosis (CAD) systems to provide more objective and reproducible image assessment [5, 6].

Recent development of artificial intelligence (AI), especially deep learning (DL), has greatly enhanced the process of analyzing medical images [7, 8]. Using CNNs enables the automatic learning of complex image features from the raw data. The performance of CNNs in different areas of medical imaging has been impressive in radiography, CT, MRI, ultrasound, and nuclear medicine [9, 10]. This has led to new possibilities for performing an automated analysis of scintigraphy images of the thyroid gland.

The feasibility of applying DL techniques to thyroid scintigraphy classification was shown in early studies. For instance, Qiao et al. [11] analyzed several CNNs for the diagnosis of thyroid scintigraphy and provided the prospect of AI-assisted classification through transfer learning and synthetic augmentation. Subsequent studies extended these approaches toward larger and multicenter datasets. Yang et al. [12] designed a dual-center DL architecture for the classification of thyroid scintigraphy uptake patterns utilizing a pre-trained CNN model and obtained excellent results on both internal and external validation. Nevertheless, reported performance in thyroid scintigraphy remains strongly dependent on dataset composition, class definitions, image-selection criteria, augmentation strategies, pretrained architectures, and validation procedures. Multicenter datasets introduce additional variability related to imaging equipment, acquisition protocols, reconstruction procedures, patient populations, and clinical practice. Such heterogeneity can result in domain shifts and may reduce the generalizability of models developed on data from a single or highly controlled center. Recent work, including

Thyroidomics by Sabouri et al. [13] has further emphasized the importance of evaluating robustness across centers and under realistic data heterogeneity.

Beyond representation learning, a more fundamental limitation of a conventional multiclass CNN is that a single feature extractor and classifier must simultaneously model all diagnostic categories, despite potentially different visual and functional characteristics among diseases. An alternative strategy is the Mixture-of-Experts (MoE) paradigm. MoE was originally introduced as a modular learning framework in which multiple expert networks specialize in different subsets or aspects of a learning problem, and their outputs are combined to generate the final decision [14]. Modern MoE architectures extend this principle through trainable routing or fusion mechanisms, allowing specialized computational pathways to contribute differently to the prediction [15].

This divide-and-specialize strategy is particularly attractive for medical imaging because different pathological categories may be characterized by distinct morphological, textural, anatomical, or functional patterns. Recent studies have begun to exploit MoE architectures in several medical-imaging domains. For example, MoE approaches have been used to integrate lesion-specific experts in mammographic breast-cancer classification, to combine global and local MRI representations for differential diagnosis, and to learn specialized representations for mammography analysis [16]. Recent work has also extended MoE-based fusion to nuclear-medicine PET/CT applications, highlighting the potential of expert specialization for heterogeneous functional and anatomical imaging data [17].

However, the potential of class-specific expert learning for thyroid scintigraphy has not been systematically investigated. In the present study, we therefore developed a class-specific MoE framework for thyroid scintigraphy, in which four CNN experts were trained to specialize in diffuse goiter, tumoral disease, thyroiditis, and normal scintigraphy. Each expert generated a class-specific logit, and the joint expert-logit vector was subsequently processed by a trainable fusion network to produce the final four-class prediction. Thus, unlike conventional sparse-routing MoE architectures, the proposed framework does not rely on selecting a single expert; rather, it exploits the complementary responses of all class-specific experts during the final diagnostic decision. Analysis of these expert logits additionally provides an interpretable view of expert specialization and their contribution to classification. In addition to predictive performance, expert-response analysis was performed to investigate whether the individual experts developed diagnostically meaningful specialization patterns. To support transparency and reproducibility, the de-identified dataset, source code, and experimental configuration are made available at <https://elmirayazdani.github.io/thyroid-scintigraphy-xai/>.

2. MATERIALS AND METHODS

2.1. Study Workflow

The overall experimental workflow of this study is summarized in **Figure 1**. The proposed pipeline was organized into six sequential stages, including multicenter data acquisition and initial screening, image preprocessing, framework-specific input construction, model development and evaluation, experimental implementation, and comprehensive model analysis.

First, retrospective [^{99m}Tc]Tc-pertechnetate thyroid scintigraphy images were collected from five nuclear medicine centers and underwent an initial quality assessment. Images affected by severe degradation, including excessive noise, motion artifacts, incomplete thyroid visualization, poor contrast, cropping, or other acquisition abnormalities, were excluded, while moderate variations were retained to preserve real-world multicenter imaging heterogeneity.

The retained images were processed using a sequential enhancement pipeline designed to reduce acquisition-related degradation while maintaining diagnostically relevant tracer uptake patterns. The preprocessing procedure included denoising, deblurring, contrast enhancement, image resizing to 128×128 pixels, intensity normalization to the range $[0,1]$, and training-only data augmentation.

Following preprocessing, framework-specific inputs were generated. The baseline CNN framework received the original single-channel grayscale scintigraphy images. The proposed MoE framework employed four class-specific expert networks corresponding to diffuse goiter, tumoral disease, thyroiditis, and normal scintigraphy. The expert outputs were subsequently integrated through a fusion network to generate the final four-class prediction.

All models were developed and evaluated under an identical experimental setup using stratified five-fold cross-validation (CV), consistent data partitioning, and the same training strategy to ensure a fair comparison between frameworks. Finally, model performance was assessed using quantitative evaluation metrics, while explainability analysis and descriptive statistical analysis of expert responses were performed to investigate prediction behavior and framework differences.

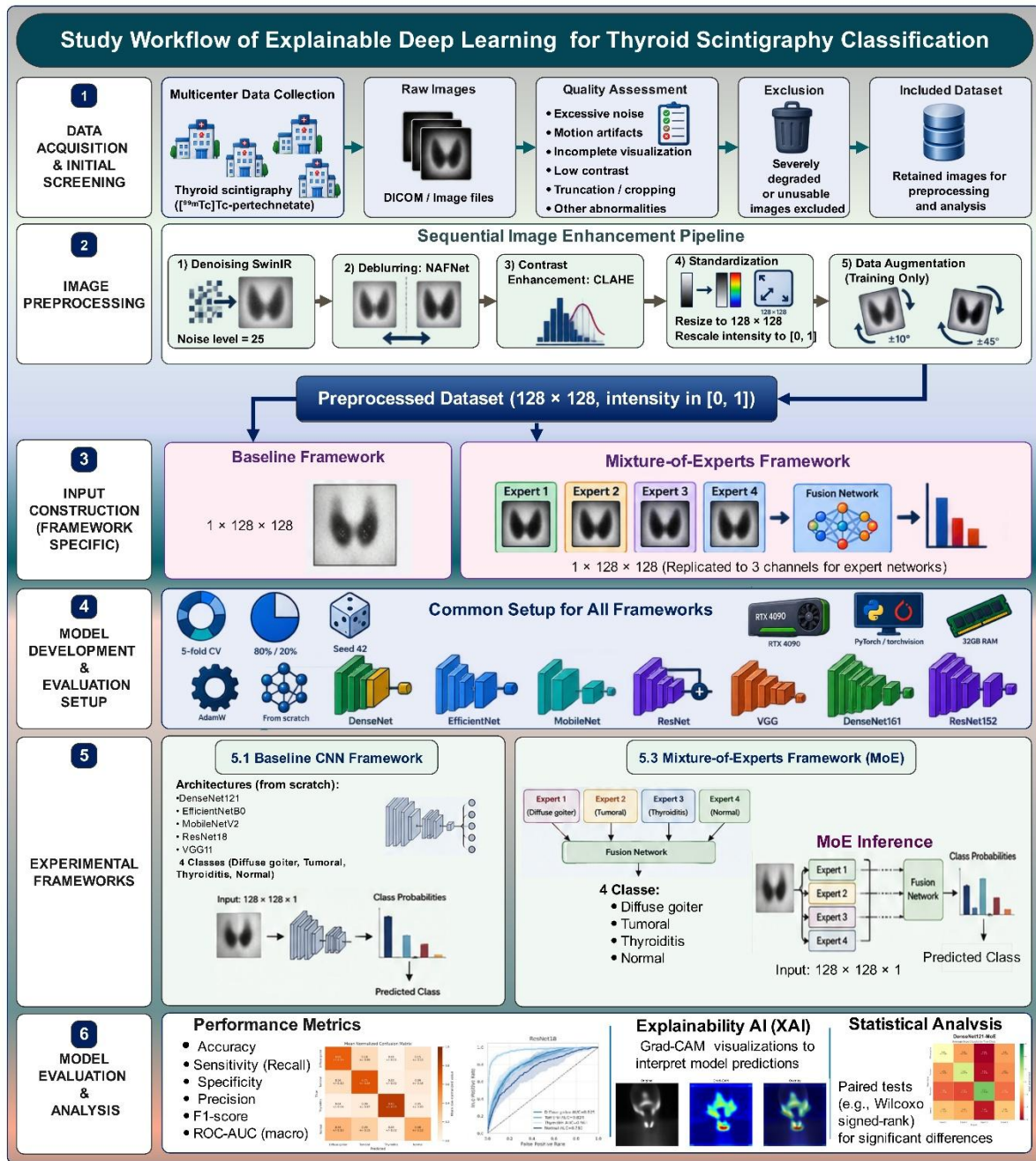


Figure 1. Overall study workflow. The workflow represents the entire experimental process, from multicenter thyroid scintigraphy data acquisition and quality check to image preprocessing, framework-specific input construction, model training with stratified five-fold cross-validation, experimental analysis of baseline CNN and mixture-of-experts (MoE) architectures, performance evaluation, interpretability assessment, and descriptive statistical analysis of expert responses.

2.2. Dataset Description

2.2.1. Data Source and Study Population

The multicenter dataset used in this study consisted of retrospective ^{99m}Tc -pertechnetate thyroid scintigraphy images collected from five nuclear medicine departments. The dataset was established to represent

real-world clinical variability by incorporating images acquired from different centers with variations in imaging systems, acquisition protocols, and patient populations.

All available thyroid scintigraphy examinations performed between June 2021 and November 2025 across the five participating centers were initially screened according to predefined inclusion and exclusion criteria. Inclusion criteria were: (1) patients who underwent [^{99m}Tc]Tc-pertechnetate thyroid scintigraphy for clinical evaluation of thyroid function or structural abnormalities; (2) availability of interpretable planar thyroid scintigraphy images with sufficient visualization of the thyroid gland; (3) confirmed clinical diagnosis based on nuclear medicine reports, laboratory findings, and available clinical information; and (4) assignment to one of the predefined diagnostic categories: diffuse goiter, tumoral disease, thyroiditis, or normal scintigraphy.

Images were excluded if they demonstrated severe quality degradation or insufficient diagnostic information, including marked motion artifacts, excessive noise, incomplete visualization of the thyroid gland, severe cropping or truncation, poor tracer contrast, corrupted image files, or inadequate acquisition quality. However, images with moderate variations in quality were retained to preserve the heterogeneity encountered in routine multicenter clinical practice.

After applying the selection criteria, a total of 2,373 thyroid scintigraphy images were included and categorized into four diagnostic groups: diffuse goiter (n = 691, 29.1%), tumoral disease (n = 922, 38.9%), thyroiditis (n = 423, 17.8%), and normal scintigraphy (n = 337, 14.2%). Demographic information was available for 1,732 participants, including 1,315 females (75.9%) and 417 males (24.1%). Among participants with valid age information, the mean age was 46.4 ± 14.5 years (range, 2–89 years).

2.2.2. Imaging Systems and Acquisition Characteristics

All thyroid scintigraphy examinations were performed using [^{99m}Tc]Tc-pertechnetate at the five participating nuclear medicine centers. Acquisition was performed using center-specific gamma-camera systems and protocols, resulting in differences in scanner manufacturer and model, collimator configuration, native image matrix, pixel size, zoom factor, and energy-window settings. The principal acquisition characteristics were obtained from the available DICOM metadata and are summarized in **Table 1**.

Table 1. Imaging systems and acquisition characteristics of the five-center thyroid scintigraphy dataset.

Center	Manufacturer/model	Collimator	Native matrix	Pixel size (mm)	Zoom	Energy window (keV)
1	Siemens Encore2, Symbia T2	LEHR, Parallel-hole	128 × 128	2.397 × 2.397	2.286	129.43–150.41
2	GE Medical Systems INFINIA	LEHR, parallel-hole	128 × 128	2.209 × 2.209	2.0	126–154
3	MiE / model not identified	LEAP, parallel-hole	128 × 128	1.125 × 1.125	1.0	119–161
4	Mediso AnyScan S-NN-3s-60R	LEHR, parallel-hole	128 × 128	1.183 × 1.183	4.0	126.45–154.55
5	Siemens IP1	LEHR, Parallel-hole	256 × 256	0.749 × 0.749	3.2	129.29–150.25

*Abbreviations: LEHR, low-energy high-resolution; LEAP, low-energy all-purpose.

The acquisition characteristics varied across centers, reflecting differences in imaging hardware and center-specific clinical protocols. In particular, the dataset incorporated variation in collimator configuration, spatial sampling, acquisition zoom, and energy-window width. The native image matrix was 128×128 at Centers 1–4 and 256×256 at Center 5. These acquisition differences were retained in the source dataset to preserve the heterogeneity inherent to the multicenter collection. Before model development, all images were standardized to a common 128×128 -pixel input format using the preprocessing procedure described in Section 2.3.

2.2.3. Data Quality and Eligibility

All collected images underwent an initial quality assessment before inclusion in the modeling dataset. Images were reviewed for technical adequacy and the presence of diagnostically interpretable thyroid information. Files with severe acquisition errors, corrupted image data, invalid dimensions, incomplete image content, or inadequate visualization of the thyroid region were excluded.

Moderate variations in image appearance, including differences in background activity, image contrast, count distribution, and acquisition characteristics, were retained when the thyroid remained diagnostically interpretable. This approach preserved variations encountered under routine clinical acquisition conditions and maintained the heterogeneous nature of the multicenter dataset. Following quality assessment, eligible images underwent the preprocessing and image-standardization procedures described in Section 2.3.

2.3. Preprocessing

2.3.1. Data Quality Assessment

Prior to preprocessing and model development, all thyroid scintigraphy images underwent a quality assessment to identify acquisition- and image-related issues that could affect subsequent analysis. Images were visually reviewed for excessive background noise, motion artifacts, incomplete thyroid visualization, low contrast, truncation, and other acquisition abnormalities. File integrity, image completeness, dimensional validity, and the presence of a diagnostically visible thyroid region were also verified.

Those with highly corrupted images with artifacts, image acquisition errors, corruption of image information, wrong image dimensions, incomplete image information, and insufficient visualization of the thyroid area were excluded from the analysis. On the other hand, images with moderate noise, blurring, contrast changes, and differences in background activity and count were included where the thyroid is still diagnostically visible. This is done to capture the variability of actual clinical image acquisition.

2.3.2. Image Enhancement

Subsequent to the quality assessment step, all images that were found to be eligible went through a chain of image enhancement methods including denoising, deblurring, and contrast enhancement. Images for thyroid scintigraphy have low spatial resolution and are prone to noise, blurriness, and contrast variation due to the inherent features of radionuclide imaging. The purpose of image enhancement was to enhance image quality while retaining important diagnostic patterns of radiotracer uptake.

First, image denoising was performed using SwinIR, a transformer-based image-restoration framework that employs shifted-window self-attention to model local and non-local image dependencies [18]. SwinIR was applied with a noise-level parameter of 25 to suppress acquisition-related noise while preserving spatial information in the scintigraphy images.

The denoised images were subsequently processed using NAFNet for deblurring [19]. NAFNet is an image-restoration architecture based on nonlinear-activation-free network blocks and was used to reduce image blur while preserving the spatial distribution of radiotracer uptake. The same deblurring configuration was applied to all images without image-specific parameter adjustment.

Finally, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied for local contrast enhancement [20]. Unlike global histogram equalization, CLAHE operates on local image regions and limits histogram amplification to control excessive enhancement of noise. A clip limit of 2.0 and a tile-grid size of 8×8 were used consistently across all images.

2.3.3. Image Standardization

After going through the image processing stage, the images underwent standardization to ensure uniformity in the input format for model training purposes. The images obtained from DICOM images were converted into JPEG image format as input for training the model. All the images obtained were standardized to the size of 128×128 pixels, ensuring uniformity in spatial size across the multi-center data and all experiments conducted while still maintaining the necessary information for classification.

The resized images were normalized to the intensity value range of $[0,1]$. The standardized 128×128 images were thus the common preprocessed data set used to construct the framework-specific inputs. In the case of the CNN baseline framework, the standardized image was used as an input. In the case of the MoE framework, the standardized image was repeated across three channels, as explained in Section 2.5.2.

2.3.4. Data Augmentation

To improve model generalizability and reduce the risk of overfitting, data augmentation techniques were applied during the training phase in each CV fold [21, 22]. The augmentation strategy was designed to simulate minor variations that may occur during image acquisition and patient positioning while preserving clinically relevant tracer uptake patterns. Specifically, random horizontal flipping was applied with a probability of 0.5, and random rotations within a range of ± 10 degrees were performed. These transformations have been considered carefully not only to improve the model's robustness to spatial variations but also to avoid unrealistic anatomical changes or altering diagnostic characteristics.

Despite these augmentation strategies, the distribution of diagnostic classes remained imbalanced after dataset partitioning, with the number of training samples per group ranging from 308 to 822 images. Such imbalance is common in medical imaging datasets and may negatively affect model optimization by biasing predictions toward majority classes and reducing sensitivity for underrepresented diagnostic categories [21, 22]. Simple oversampling methods based on image duplication can increase class frequency but do not provide additional image diversity and may increase the risk of overfitting to limited minority-class appearances.

As an alternative capable of introducing genuinely new, if synthetic, training examples, a class-conditional generative adversarial network (GAN) was trained to synthesize additional scintigraphy images for each diagnostic group, consistent with prior work applying GAN-based image synthesis to expand small, imbalanced medical imaging datasets [23-26]. In this study, a single generator-discriminator pair was trained jointly across all four groups, conditioned on the group label, rather than training four separate group-specific GANs: with as few as 308 real images available for the smallest group, an independently trained generator for that group alone would have had too little data to learn a stable representation, whereas pooling all groups allowed the shared convolutional layers to learn general scintigraphy texture and radiotracer-uptake statistics from the full training set while a class embedding steered which diagnostic pattern was produced.

Both networks followed a deep convolutional GAN (DCGAN) design, adapted for single-channel 128×128 inputs and for conditioning on the group label [27]. The generator G mapped a 128-dimensional latent vector $z \sim \mathcal{N}(0, I)$, concatenated with a 50-dimensional learned embedding of the group label y , through five transposed-convolutional blocks, each followed by batch normalization and a ReLU activation, that progressively upsampled the representation from a single spatial position to the full 128×128 resolution, ending in a final transposed-convolutional layer with a hyperbolic-tangent activation that constrained pixel intensities to $[-1, 1]$. The discriminator D mirrored this structure: the candidate image, real or generated, was concatenated with a spatial map obtained by projecting the same label embedding across the full 128×128 grid, and the resulting two-channel input was passed through five spectrally normalized convolutional layers with LeakyReLU activations (negative slope 0.2), followed by a final convolutional layer that collapsed the representation to a single scalar realism score $D(x, y)$ [28]. Spectral normalization constrains the Lipschitz constant of each discriminator layer and was applied throughout D to stabilize adversarial training given the small batch sizes and limited per-class sample counts available for this dataset.

Both networks were optimized with the hinge adversarial loss, which in practice produced steadier gradient behavior than the original two-class softmax cross-entropy GAN objective at this data scale [29]:

$$\begin{aligned}\mathcal{L}_D &= \mathbb{E}_{(x, y) \sim p_{\text{data}}} [\max(0, 1 - D(x, y))] + \mathbb{E}_{z \sim \mathcal{N}(0, I), y} [\max(0, 1 + D(G(z, y), y))], \\ \mathcal{L}_G &= -\mathbb{E}_{z \sim \mathcal{N}(0, I), y} [D(G(z, y), y)].\end{aligned}\tag{1}$$

where $\mathcal{L}_D$ and $\mathcal{L}_G$ denote the discriminator and generator losses, respectively. In these formulations, x represents a real thyroid scintigraphy image sampled from the training distribution p_{data} , y denotes the corresponding diagnostic class label, z represents a latent vector randomly sampled from a normal distribution $\mathcal{N}(0, I)$, and $G(z, y)$ indicates the synthetic image generated by the conditional generator based on the latent representation and class condition. The discriminator function $D(x, y)$ estimates the realism score of an image x conditioned on class label y , whereas $D(G(z, y), y)$ represents the discriminator response for a generated image. The expectation operators $\mathbb{E}$ calculate the average loss over real and generated image distributions. The hinge terms enforce a margin-based separation between real and synthetic samples, encouraging the discriminator to assign high scores to real images and low scores to generated images while providing stable gradients for generator optimization.

With as few as 308 real images available for one group, a discriminator trained directly on the raw images can memorize the training set within a small number of epochs and stop providing the generator with a useful

learning signal. To delay this, the same differentiable, randomly sampled augmentation $T(\cdot)$, composed of brightness and contrast jitter, small spatial translations, and cutout, was applied to both real and generated images immediately before they reached the discriminator [30], so that Equation (1) was in practice evaluated on $T(x)$ and $T(G(z, y))$ rather than on the raw images. Because T is differentiable, gradients computed from the augmented discriminator output still propagated back through T into the generator. This kept the discriminator from memorizing the small pool of real images too quickly, without changing the underlying image distribution the generator was learning to approximate.

During an initial training run, the generator was observed to converge toward producing near-identical images for a given group regardless of the latent code z supplied to it, a form of mode collapse that can arise when small per-class sample counts leave the generator little incentive to preserve latent diversity once it has found a single output capable of fooling the discriminator. Visual inspection of individual generated images was not sufficient to catch this. Each output could still look like a plausible scintigraphy pattern on its own; the loss of variation became apparent only when several outputs from the same group were compared side by side. To directly counteract this failure mode, a mode-seeking regularization term was added to the generator loss, computed on a second, independently sampled pair of latent codes $z_1, z_2 \sim \mathcal{N}(0, I)$ mapped to the same group label y [31]:

$$\begin{aligned}\mathcal{L}_{ms} &= \mathbb{E}_{z_1, z_2 \sim \mathcal{N}(0, I), y} \left[\frac{\|z_1 - z_2\|_1}{\|G(z_1, y) - G(z_2, y)\|_1 + \epsilon} \right], & \epsilon &= 10^{-5}, \\ \mathcal{L}_G^{\text{total}} &= \mathcal{L}_G + \lambda_{ms} \mathcal{L}_{ms}, & \lambda_{ms} &= 1.\end{aligned}\tag{2}$$

where $\mathcal{L}_{ms}$ represents the mode-seeking regularization term introduced to encourage diversity among generated images. The numerator $\|z_1 - z_2\|_1$ measures the L_1 -distance between two independently sampled latent vectors, while the denominator $\|G(z_1, y) - G(z_2, y)\|_1$ represents the corresponding L_1 -distance between the generated images conditioned on the same diagnostic class label y . The objective of this ratio is to maximize the variation in generated images relative to the variation in latent space, preventing different latent codes from collapsing into similar outputs. ϵ is a small stability constant (10^{-5}) added to the denominator to avoid numerical instability during optimization.

The total generator objective $\mathcal{L}_G^{\text{total}}$ combines the original adversarial generator loss $\mathcal{L}_G$ with the mode-seeking regularization term weighted by λ_{ms} . In this study, $\lambda_{ms} = 1$, indicating that the diversity constraint contributed equally with the adversarial objective during generator optimization. The latent vectors z_1 and z_2 were independently sampled from a standard normal distribution $\mathcal{N}(0, I)$, and both were conditioned on the same diagnostic class label y to ensure that the regularization promoted intra-class diversity rather than changes between diagnostic categories.

Minimizing the ratio in Equation 2 penalizes the generator whenever two dissimilar latent codes yield similar images for the same group, directly rewarding it for letting z continue to influence its output rather than collapsing to a single canonical appearance per group.

To confirm this regularization was effective in practice, generator diversity was monitored quantitatively rather than by visual inspection alone. For a batch of n images sharing a group label, diversity was measured as the mean pairwise L_1 pixel distance,

$$\text{Div}(\mathcal{X}) = \binom{n}{2}^{-1} \sum_{i < j} \frac{1}{HW} \|x_i - x_j\|_1, \quad x_i, x_j \in \mathcal{X}, \quad (3)$$

where $\text{Div}(X)$ denotes the diversity score of a generated image set X , calculated as the average pairwise L_1 distance among all generated images within the same diagnostic category. In this formulation, n represents the number of generated images in the evaluated batch, and $\binom{n}{2}$ indicates the total number of unique image pairs considered for diversity estimation. The term x_i and x_j represent two generated images sampled from the same diagnostic class, while H and W denote the image height and width, respectively. The normalization factor $1/(HW)$ converts the accumulated pixel-wise distance into the average difference per pixel. The L_1 -norm $\|x_i - x_j\|_1$ measures the absolute pixel intensity difference between two generated images.

Diversity was evaluated every 25 training epochs using freshly generated samples from each diagnostic group and was normalized relative to the corresponding diversity measured from real training images. A diversity ratio close to or greater than one indicated that synthetic images exhibited variability comparable to real scintigraphy images, whereas values approaching zero suggested reduced diversity and potential mode collapse. The final GAN checkpoint used for synthetic image generation was selected based on this diversity criterion; checkpoints in which any diagnostic group showed a diversity ratio below 15% of the corresponding real-data baseline were rejected in favor of an earlier stable checkpoint. This quality-control strategy ensured that synthetic images introduced meaningful variability rather than near-duplicate appearances. The final GAN training configuration is summarized in **Table 2**.

Table 2. Training configuration for the class-conditional GAN used to synthesize additional training images for each diagnostic group.

Hyperparameter	Value
Epochs	800
Batch size	32
Latent dimension	128
Group-embedding dimension	50
Optimizer	Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$)
Generator/discriminator learning rate	$2 \times 10^{-4} / 2 \times 10^{-4}$
Adversarial loss	Hinge loss
Differentiable augmentation policy	Color jitter, translation, cutout
Mode-seeking loss weight (λ_{ms})	1.0
Diversity check interval	Every 25 epochs
Diversity acceptance threshold	$\geq 15\%$ of real-data diversity
Target images per group (post-augmentation)	822 (majority group size)
Random seed	42

For each diagnostic group, the trained generator was used to synthesize enough new images to bring that group’s training count up to the size of the largest group, 822 images, matching the target already used by the duplication-based oversampling strategy so that the two class-balancing approaches produced training sets of identical size and remained directly comparable. Each synthetic image was assigned one of the original raw diagnostic labels subsumed by its group, sampled in the same proportion observed among the real images of that group, so that the label composition within each group mirrored that of the real data rather than treating the four broad groups as internally homogeneous. The generator was trained exclusively on the training partition of the dataset; the held-out test images were never used to train it. Consistent with the augmentation policy described

above, synthetic images were added only to the training set, while validation and test images remained entirely real throughout this study.

Data augmentation, comprising both the geometric transformations described above and the GAN-based synthesis used for class balancing, was applied exclusively to the training set, while validation and test images remained unchanged to ensure an unbiased evaluation of model performance.

2.4. Experimental Setup

Two experimental frameworks were implemented to evaluate DL approaches for thyroid scintigraphy classification [8]. The evaluation includes a baseline framework built on standard CNN architectures and an MoE framework that decomposes classification across class-specific experts.

The baseline framework employed established CNN architectures trained directly on the preprocessed scintigraphy images. This framework served as the reference configuration for assessing the alternative input representation and classification task investigated in the subsequent experiments [8]. The MoE framework formulated the four-class classification task as four class-specific binary classification problems, each assigned to a separate CNN-based expert network. Three MoE configurations were evaluated using DenseNet121, MobileNetV2, and VGG11 as the expert backbone architectures. Within each configuration, four independent experts were constructed, corresponding to diffuse goiter, tumoral disease, thyroiditis, and normal scintigraphy. For each target class, the corresponding expert was trained to distinguish samples belonging to that class from those belonging to the remaining three classes. The outputs of the four experts were subsequently combined by a trainable fusion network to produce the final four-class prediction. This architecture follows the general MoE principle of using specialized sub-models for distinct components of a learning problem, while the fusion stage adopts the stacked-generalization principle of training a second-level model on the outputs of multiple base models.

Section 2.5 describes the dataset partitioning, CV protocol, training procedures, checkpoint-selection strategy, and computational environment common to the two frameworks. Framework-specific architectures, input construction procedures, training configurations, and inference strategies are detailed in the following subsections.

2.5. Model Development

Both experimental frameworks followed a common dataset-partitioning, CV, checkpoint-selection, and computational protocol. These shared procedures are described below, while framework-specific architectures and training configurations are presented in the corresponding subsections.

Model evaluation was performed using stratified five-fold CV. Data partitioning was performed at the patient level, such that all images belonging to the same patient were restricted to a single fold. The dataset was partitioned into five folds according to the original four-class diagnostic labels using a fixed random seed of 42. For each CV iteration, four folds, corresponding to approximately 80% of the dataset, were used for training, while the remaining fold, approximately 20%, was used for validation. This patient-level partitioning prevented images from the same individual from appearing in both the training and evaluation partitions. The same fold

assignments were retained across all experimental frameworks and model architectures. Within each fold, the model checkpoint achieving the highest validation accuracy was retained for subsequent evaluation.

AdamW was used as the optimizer throughout the study with a weight decay of 1×10^{-4} (Loshchilov and Hutter 2019). Multiclass models were optimized using cross-entropy loss without class weighting, while the class-specific experts in the MoE framework used the two-class cross-entropy formulation described in Section 2.5.2. Framework-specific learning rates, batch sizes, learning-rate schedules, training durations, and other optimization settings are reported in their respective subsections.

Every convolutional backbone across all three frameworks, including the class-specific experts and the fusion feature extractor of the MOE framework, was initialized from ImageNet-pretrained weights rather than trained from scratch. Whether natural-image pretraining benefits medical imaging tasks is not settled, given the substantial domain shift between natural photographs and thyroid scintigraphy in resolution, intensity distribution, and visual statistics); nonetheless, fine-tuning pretrained convolutional networks has repeatedly been shown to match or exceed training the same architectures from scratch on modestly sized medical imaging datasets, while converging in fewer epochs. Given the size of the dataset used in this study, pretrained initialization was adopted across the two frameworks. Because standard torchvision implementations of these architectures expect three-channel input, every single-channel grayscale image was mapped onto a three-channel input; the specific channel-construction strategy, either naive replication of the grayscale channel or a composite of derived representations, is described separately for each framework below.

All frameworks operated on images standardized to 128×128 pixels as described in Section 2.3 and addressed the same four-class classification task. Model implementation and training were performed using PyTorch and torchvision on a single NVIDIA GeForce RTX 4090 GPU with 24 GB of VRAM and 32 GB of system RAM.

2.5.1. Baseline CNN Framework

The baseline framework includes five established CNN architectures: DenseNet121, EfficientNet-B0, MobileNetV2, ResNet18, and VGG11. These architectures were selected because they are standard image-classification benchmarks that are frequently used in medical imaging research, and every model was initialized from ImageNet-pretrained weights. These architectures represent different CNN design families and were included to provide a common reference for the alternative input representation and classification frameworks investigated in the subsequent experiments. Each single-channel grayscale image was mapped onto the three-channel input expected by these architectures through naive channel replication, duplicating the grayscale image across all three input channels, and the resulting image was normalized using the standard ImageNet channel-wise mean and standard deviation rather than statistics computed from this dataset, consistent with reusing off-the-shelf pretrained weights.

Each architecture was trained for a maximum of 100 epochs per fold using the same baseline-specific training configuration. The initial learning rate was 1×10^{-3} , with a batch size of 32 and a cosine-annealing learning-rate schedule (CosineAnnealingLR; $T_{\max} = 100$, $\eta_{\min} = 10^{-6}$). The models received images as input. The general augmentation protocol described in Section 2.3.4, comprising random horizontal flipping ($p = 0.5$) and rotation within $\pm 10^\circ$, was applied only to the training partition. Data loading was implemented through four workers in

parallel. Such options were chosen to create equal conditions for training of all the studied CNNs according to the traditional experiment protocol.

2.5.2. Mixture-of-Experts (MoE) Framework

The third experimental framework tested whether decomposing the four-class problem into four class-specific binary sub-problems, each handled by its own expert, improves discrimination relative to a single network that must separate all four categories directly. Rather than training one classifier to assign a scintigraphy image to one of four categories at once, one expert was trained per class to answer a narrower question: does this image belong to my assigned class or not? A trainable fusion network then combined the four experts' outputs, together with an image-level feature representation, into a single four-class prediction. This design follows the general mixture-of-experts principle of assigning specialized sub-models to partitions of a problem, and the fusion stage follows the stacked-generalization approach of training a second-level model on the outputs of a set of base models. The framework was evaluated as three independent, self-contained pipelines, each built around one of the same three pretrained backbones: DenseNet121, MobileNetV2, and VGG11. Within a given pipeline, the same backbone architecture supplied all four experts and the fusion network's image-feature extractor, though each of these five components was trained with its own independent set of weights.

Building on the common input format, each 128×128 image was replicated across three identical channels to match the input dimensionality required by the DenseNet121, MobileNetV2, and VGG11 expert backbones. Each channel was normalized with the fixed ImageNet channel-wise mean and SD rather than statistics computed from this dataset, since fixed ImageNet normalization is expected by the pretrained weights used throughout this framework.

Each of the four experts, one per class, replaced the lightweight, custom convolutional architecture used in an earlier version of this experiment with the pretrained backbone described above: for a given pipeline, every expert's original classification layer was replaced with a single linear layer mapping directly to two output logits, with no additional hidden layers or dropout between the backbone's pooled feature representation and the output, unlike the dropout-plus-linear head used in the three-channel framework. This substitution of a validated, pretrained backbone for the earlier custom architecture removed a confound from the expert-versus-single-classifier comparison motivating this framework, since the experts were now built from the same building blocks already evaluated in the other two frameworks rather than from a bespoke small network. For expert c , the training label for an image with grouped label y was defined as:

$$y_c = \begin{cases} 1, & y = c \\ 0, & y \neq c \end{cases} \quad (4)$$

so that every expert was trained as a one-versus-rest binary classifier, with the three non-target classes pooled into a single negative class.

Writing $X \in \mathbb{R}^{3 \times 128 \times 128}$ for the input to a given expert and $\phi_b^{(c)}(X) \in \mathbb{R}^{d_b}$ for the pooled feature representation produced by expert c 's own copy of pretrained backbone b , that is, by its convolutional layers and, for VGG11, its fully connected layers up to but excluding the final classification layer, with embedding dimension d_b equal

to 1024 for DenseNet121, 1280 for MobileNetV2, and 4096 for VGG11, each expert’s replacement linear layer mapped this representation directly to a pair of output logits,

$$z_c(X) = W_c \phi_b^{(c)}(X) + b_c \in \mathbb{R}^2, \quad W_c \in \mathbb{R}^{2 \times d_b}, b_c \in \mathbb{R}^2, \quad (5)$$

where W_c and b_c are expert c ’s linear-layer weights and bias. The positive-class logit used downstream, corresponding to the “belongs to class c ” output, was the second component of this pair, $s_c(X) = [z_c(X)]_1$. Each expert was trained independently to minimize the two-class cross-entropy loss:

$$\mathcal{L}_c = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp([z_c(X^{(i)})]_{y_c^{(i)}})}{\sum_{k \in \{0,1\}} \exp([z_c(X^{(i)})]_k)}, \quad (6)$$

with $y_c^{(i)}$ defined by Equation (4) for training image $X^{(i)}$.

Positive and negative samples were imbalanced for every expert, since each target class made up a minority of the training partition relative to the pooled remainder. This imbalance was addressed the same way as in the earlier version of this experiment: the minority class in each expert’s training set was resampled with replacement up to the size of the majority class, and a bounded random rotation of up to $\pm 45^\circ$ was applied to the resampled, duplicate instances. As before, this balancing and augmentation procedure was applied only to the training partition of each expert and differed from the general augmentation strategy described in Section 2.3.4; horizontal flipping was not applied. This per-expert balancing addresses a different imbalance axis from the four-way class balancing performed by the GAN synthesis, and remained necessary even though the training pool as a whole was already balanced across the four diagnostic groups. Validation data for each expert used the original, unbalanced class distribution so that reported validation accuracy reflected the true prevalence of the target class.

Training proceeded in two stages within each fold, repeated independently for each of the three backbones, as shown in Figure 2. In the first stage, the four experts were trained independently, each with the AdamW optimizer, gradient-norm clipping, and a plateau-based learning-rate schedule, as summarized in **Table 3**. In the second stage, all four experts were frozen, and a fusion network built around the same backbone, used here purely as an image-feature extractor with its own classification head discarded, was trained to combine the frozen experts’ outputs with this image-level embedding into the final four-class prediction. For each training image, every frozen expert produced a single scalar: the raw output logit of its positive class, prior to any softmax or sigmoid normalization, which was used directly as a feature rather than as a calibrated probability.

Table 3. Training configuration specific to each of the four class-specific experts, repeated independently for each backbone (see Table 2 for settings shared across all frameworks).

Hyperparameter	Value
Epochs	100
Batch size	16
Learning rate	1×10^{-4}
Learning rate scheduler	ReduceLROnPlateau (mode: min, factor = 0.5, patience = 5)
Gradient norm clipping	Max norm = 1.0 (DenseNet121, MobileNetV2); = 2.0 (VGG11)
Loss function	Binary cross-entropy (one-versus-rest)

Class balancing	Random oversampling to majority class size (training set only)
Pretrained weights	ImageNet-1k
Input channels	3 (grayscale replicated across channels)
Number of experts	4 (one per class)
Number of workers	2

Table 4 summarizes the fusion-stage training configuration. Both stages followed the common CV and checkpoint-selection protocol described in Section 2.1.

Table 4. Training configuration specific to the fusion network, repeated independently for each backbone (see Table 2 for settings shared across all frameworks).

Hyperparameter	Value
Epochs	50
Batch size	16
Learning rate	1×10^{-3} (DenseNet121, MobileNetV2); 1×10^{-4} (VGG11)
Learning rate scheduler	ReduceLROnPlateau (mode: min, factor = 0.5, patience = 5)
Gradient norm clipping	Max norm = 1.0 (DenseNet121, MobileNetV2); = 2.0 (VGG11)
Input features	Backbone image embedding (d_b dimensions) concatenated with 4 frozen expert logits
Expert parameters	Frozen
Fusion feature-extractor parameters	Pretrained initialization, fine-tuned jointly with the fusion head
Data balancing	None (original class distribution)

Figure 2 summarizes the resulting architecture and inference pipeline. The same input image was evaluated independently by all four frozen experts, and their positive-class logits were concatenated into a single four-dimensional vector,

$$v(X) = [s_0(X) s_1(X) s_2(X) s_3(X)]^T \in \mathbb{R}^4 \quad (7)$$

Unlike the earlier version of this framework, in which the fusion network’s only input was $v(X)$, the same image was also passed through the fusion network’s own copy of the backbone b , parameterized independently of the four experts’, to obtain an image-level embedding $\phi_b(X) \in \mathbb{R}^{d_b}$, using the same per-backbone embedding dimension d_b defined above. The expert logits and image embedding were concatenated into a single vector $u(X) = [\phi_b(X)^T, v(X)^T]^T \in \mathbb{R}^{d_b+4}$, which was passed to the fusion network’s two-hidden-layer head, $\text{MLP}_{\text{fusion}}: \mathbb{R}^{d_b+4} \rightarrow \mathbb{R}^{256} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^4$, with a ReLU activation and dropout at $p = 0.3$ after each of the two hidden layers, producing

$$o(X) = \text{MLP}_{\text{fusion}}(s(X)) \in \mathbb{R}^4, \quad (8)$$

The fusion network was trained to minimize the four-class cross-entropy loss:

$$\mathcal{L}_{\text{fusion}} = -\frac{1}{N} \sum_{i=1}^N \log \left[\frac{\exp([o(X^{(i)})]_{y^{(i)}})}{\sum_{k=0}^3 \exp([o(X^{(i)})]_k)} \right], \quad (9)$$

where $y^{(i)} \in \{0,1,2,3\}$ is the original grouped label of the training image $X^{(i)}$. Only the four experts’ parameters were held fixed during this stage; the fusion network’s own feature extractor, although initialized from the same ImageNet-pretrained weights, was updated jointly with the fusion MLP rather than frozen. At inference, the predicted class was the argmax of the fusion network’s output,

$$\hat{y}(X) = \arg \max_{k \in \{0,1,2,3\}} [o(X)]_k, \quad (10)$$

in the same manner as the softmax-based decision rule used in the baseline and three-channel experiments, since the fusion network’s four output units play the role that the final classification layer plays in those experiments. Because both $u(X)$ and $o(X)$ are continuous-valued, exact numerical ties were effectively improbable, and no explicit tie-breaking rule was implemented. This aggregation mechanism is a learned form of expert aggregation rather than a fixed rule such as selecting the class of the single most confident expert: the fusion network combines a global image representation with the four experts’ class-specific opinions, rather than relying on the expert logits alone as in the earlier version of this framework.

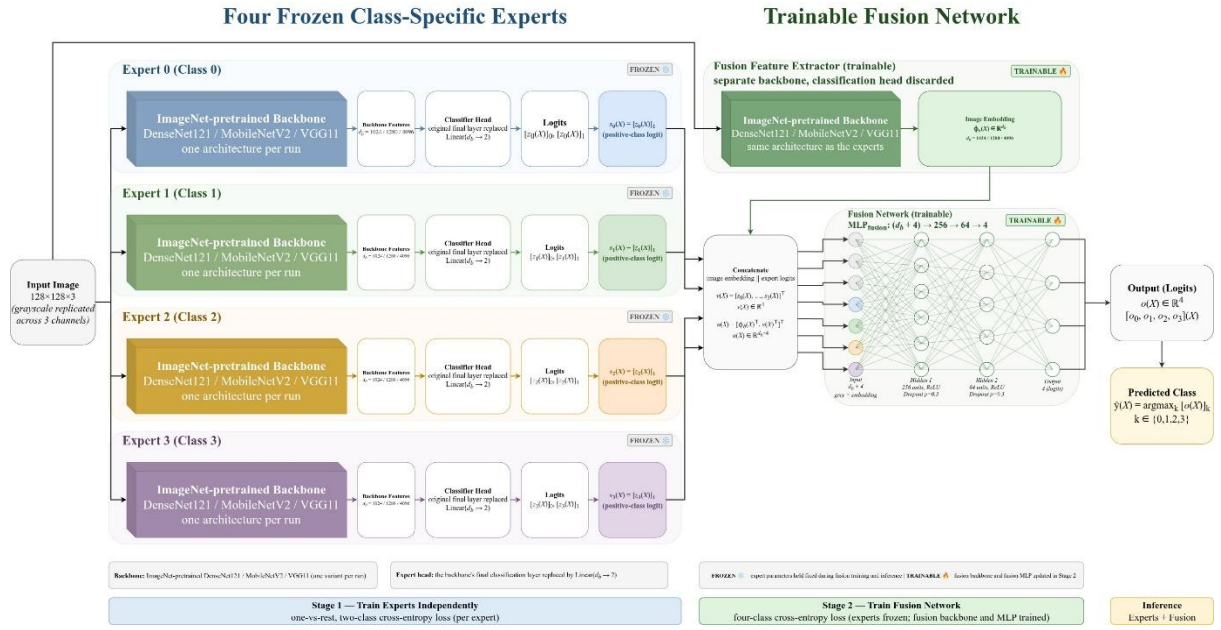


Figure 2. Architecture and two-stage training procedure of the mixture-of-experts (MoE) framework. The same $3 \times 128 \times 128$ input X obtained by replicating the preprocessed grayscale scintigraphy image across three channels, is presented both to the four class-specific experts and to the fusion network’s own image-feature extractor. Each expert is an ImageNet-pretrained backbone whose original classification layer has been replaced by a single linear layer mapping the d_b -dimensional pooled feature representation to two logits; its positive-class logit $s_c(X)$ indicates how strongly that expert favors its assigned class over the pooled remaining three. The four positive-class logits form the vector $v(X) \in \mathbb{R}^4$, which is concatenated with the image embedding $\phi_b(X) \in \mathbb{R}^{d_b}$ to give $u(X) \in \mathbb{R}^{d_b+4}$, the input to the fusion network’s two-hidden-layer classification head. The predicted class is taken as the argmax of the four-dimensional fusion output $o(X)$. Within each cross-validation fold, training proceeded in two stages: the four experts were first trained independently under a one-versus-rest, two-class cross-entropy objective (Stage 1), after which their parameters were held fixed while the fusion network, comprising both its feature extractor and its classification head, was trained with a four-class cross-entropy loss (Stage 2). A single backbone architecture, DenseNet121, MobileNetV2, or VGG11, supplies all five components of a given pipeline, with embedding dimensions d_b equal to 1024, 1280, and 4096, respectively; each component was trained with its own independent set of weights.

2.6. Classification Granularity Analysis

A binary classification task was conducted to test the hypothesis that the reduction in the number of classification categories would lead to better classification performance. The initial four-category classification task was changed to a simpler clinical binary classification task by categorizing the cases into either tumoral or

non-tumoral classes. The tumoral category contained cases categorized as tumoral in the initial set, while the non-tumoral category contained the rest, that is, diffuse goiter, thyroiditis, and normal.

All other aspects of the experiment remained the same in order to make a direct comparison between the initial four-category classification task and the new binary task.

3. RESULTS

3.1. Dataset Distribution

Table 5 summarizes the characteristics of the final dataset used for model development and evaluation. The dataset consisted of 2,373 original thyroid scintigraphy images distributed among four diagnostic categories: diffuse goiter, tumoral disease, thyroiditis, and normal scintigraphy. Among these classes, tumoral cases represented the largest proportion of the dataset (38.9%, $n = 922$), followed by diffuse goiter (29.1%, $n = 691$), thyroiditis (17.8%, $n = 423$), and normal cases (14.2%, $n = 337$).

The contribution of individual centers varied, reflecting the heterogeneous nature of the multicenter dataset. Center 1 contributed the largest proportion of images (1,022 cases; 43.1%), followed by Center 2 (606 cases; 25.5%), Center 3 (478 cases; 20.1%), Center 4 (174 cases; 7.3%), and Center 5 (93 cases; 3.9%). All four diagnostic categories were represented across all centers, although the number of available cases differed among institutions. This distribution demonstrates the multicenter variability incorporated into the dataset and provides a more realistic evaluation setting for assessing model generalizability.

The original dataset showed considerable class imbalance, with thyroiditis and normal classes having fewer samples than the largest class (tumoral disease), resulting in imbalance ratios of $2.74\times$ and $2.05\times$, respectively. To mitigate this imbalance during model training, a GAN-based augmentation strategy was applied to increase the representation of minority classes. A total of 3,288 synthetic images were generated for the training corpus, resulting in a balanced GAN training distribution in which each diagnostic class contributed 25.0% of the generated training samples. The final augmented corpus, including original and GAN-generated images, contained 3,526 images.

Table 5. Distribution of the multicenter thyroid scintigraphy dataset, class imbalance characteristics, and GAN-based augmentation strategy.

Imaging Center	Diffuse Goiter (Group 1)	Tumoral (Group 2)	Thyroiditis (Group 3)	Normal (Group 4)	Total Cases (n)	Dataset (%)
Center 1	314	468	143	97	1,022	43.1%
Center 2	147	198	121	140	606	25.5%
Center 3	130	182	93	73	478	20.1%
Center 4	69	45	42	18	174	7.3%
Center 5	31	29	24	9	93	3.9%
All Centers (n)	691	922	423	337	2,373	100.0%
Main Dataset (%)	29.1%	38.9%	17.8%	14.2%	—	—
Imbalance Ratio (vs. Smallest Class)	$2.05\times$	$2.74\times$	$1.26\times$	$1.00\times$	—	—
GAN Training Images (n)	822	822	822	822	3,288	100.0%
Final GAN Corpus (Train + Test)	894	922	859	851	3,526	—

3.2. Baseline CNN Classification Performance

The baseline experiment evaluated five established CNN architectures, VGG11, MobileNetV2, ResNet18, EfficientNetB0, and DenseNet121, using thyroid scintigraphy images under the common stratified five-fold CV protocol. All models were trained from scratch using the baseline-specific configuration described in Section 2.5.1.

Table 6 summarizes the overall performance of the five baseline CNN architectures across the five-fold CV procedure, with the best value for each performance metric highlighted in bold. DenseNet121 achieved the highest mean accuracy (0.674 ± 0.023), whereas MobileNetV2 showed the highest macro-precision (0.661 ± 0.026), macro-recall (0.644 ± 0.024), macro-F1 score (0.640 ± 0.014), and mean AUC (0.853 ± 0.016). MobileNetV2 also demonstrated comparatively low fold-to-fold variability. ResNet18 showed the lowest macro-F1 score (0.615 ± 0.057) and the greatest variability in recall and F1 score. Overall, however, performance differences among the CNN architectures were modest.

Table 6. Overall classification performance of the baseline CNN architectures across five-fold cross-validation, ordered by increasing accuracy. Values are presented as mean $\pm$ standard deviation (SD) across five-fold cross-validation.

Model	Accuracy	Precision	Recall	F1-score	AUC	Cohen's κ	MCC
VGG11	0.664 ± 0.027	0.648 ± 0.026	0.635 ± 0.027	0.633 ± 0.022	0.839 ± 0.025	0.525 ± 0.038	0.529 ± 0.039
DenseNet121	0.674 ± 0.023	0.654 ± 0.022	0.637 ± 0.023	0.636 ± 0.025	0.844 ± 0.015	0.539 ± 0.031	0.543 ± 0.030
ResNet18	0.658 ± 0.024	0.651 ± 0.026	0.621 ± 0.055	0.615 ± 0.057	0.840 ± 0.023	0.513 ± 0.046	0.519 ± 0.043
MobileNetV2	0.672 ± 0.012	0.661 ± 0.026	0.644 ± 0.024	0.640 ± 0.014	0.853 ± 0.016	0.538 ± 0.015	0.542 ± 0.014
EfficientNetB0	0.662 ± 0.014	0.654 ± 0.028	0.632 ± 0.029	0.633 ± 0.023	0.839 ± 0.023	0.523 ± 0.025	0.528 ± 0.025

*Note: Precision, recall, F1-score, and ROC-AUC are macro-averaged across the four diagnostic classes.

*Abbreviations: AUC, area under the receiver operating characteristic curve; CNN, convolutional neural network; F1-score, harmonic mean of precision and recall; κ , Cohen's kappa coefficient; MCC, Matthews correlation coefficient.

Class-specific ROC analysis (**Figure 3**) showed that thyroiditis was the most readily discriminated category for all architectures, with AUCs ranging from 0.943 to 0.961. In contrast, the normal class consistently showed the lowest AUC values (0.750–0.784). MobileNetV2 provided the highest discrimination for diffuse goiter and normal scans, while ResNet18 yielded the highest thyroiditis AUC.

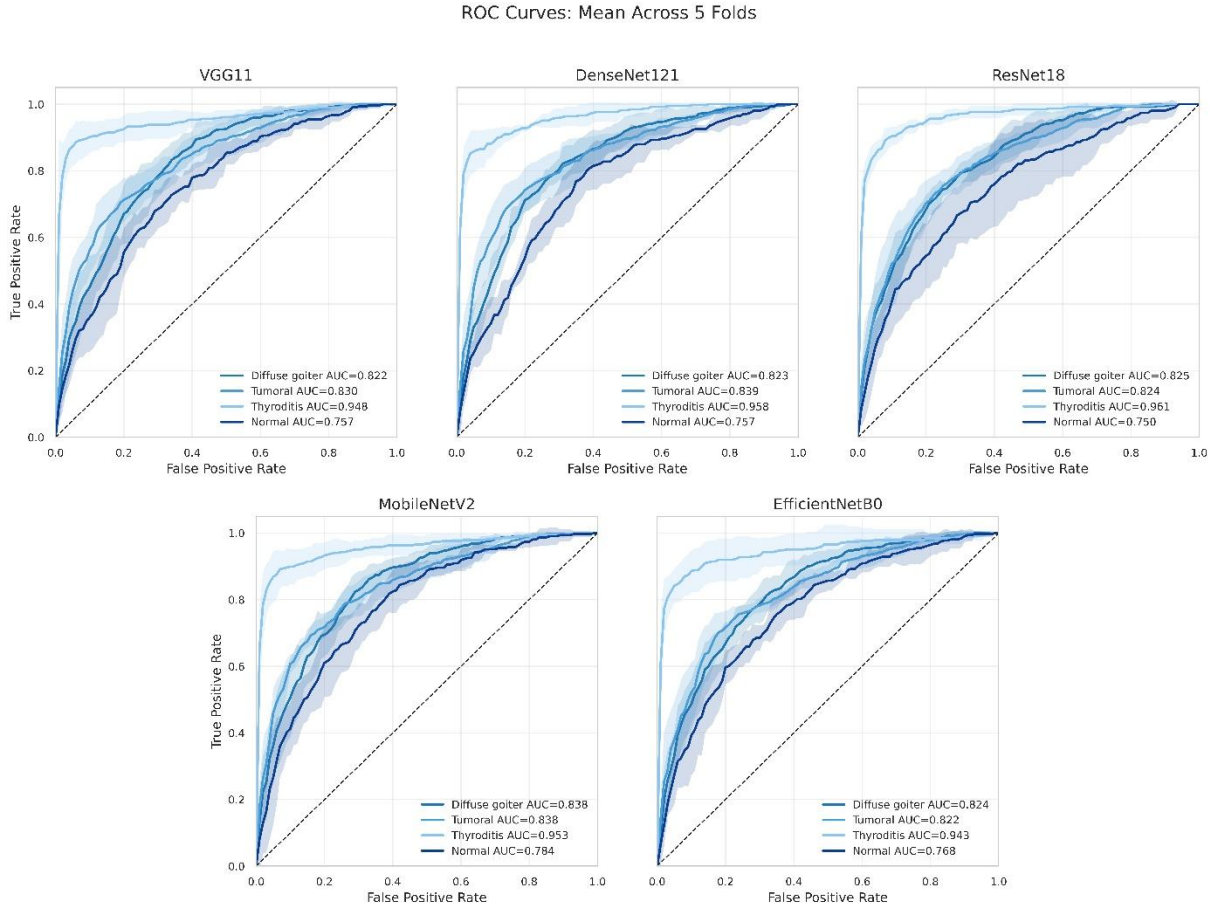


Figure 3. Class-specific receiver operating characteristic (ROC) curves for the five baseline CNN architectures across five-fold cross-validation. Each panel shows one-vs-rest ROC curves for diffuse goiter, tumoral disease, thyroiditis, and normal classes, with the corresponding mean area under the curve (AUC) reported for each class. Shaded regions indicate variability across the five folds.

Confusion-matrix analysis (**Figure 4**) revealed consistent class-specific patterns across all five CNN architectures. Thyroiditis was the most reliably classified category, with normalized diagonal values ranging from 0.82 to 0.87, followed by the tumoral class with correct classification rates of approximately 0.70–0.74. Diffuse goiter showed intermediate performance, with sensitivities ranging from 0.62 to 0.70 across models. In contrast, the normal class was the most challenging, with only 0.29–0.33 of cases correctly identified. Most normal cases were misclassified as diffuse goiter (0.35–0.42) or tumoral lesions (0.22–0.31). Among the evaluated models, DenseNet121 achieved the highest correct classification rate for diffuse goiter (0.70), ResNet18 for tumoral cases (0.74), VGG11 for thyroiditis (0.87), and MobileNetV2 for the normal class (0.33). Overall, the confusion matrices indicate that the principal source of classification error was the overlap of normal scintigraphy with diffuse goiter and tumoral patterns.

Confusion Matrices: Mean +/- 95% CI Across 5 Folds

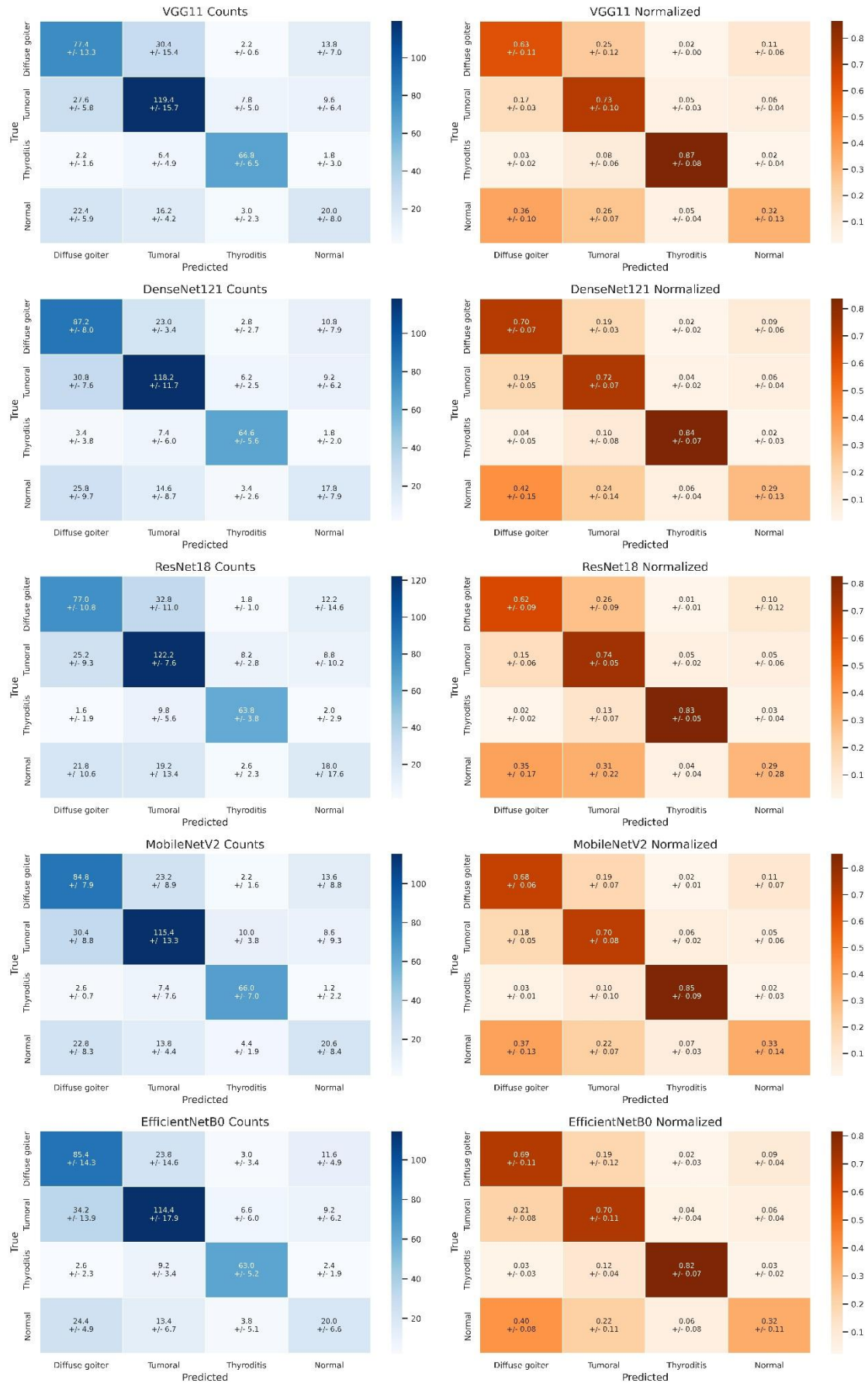


Figure 4. Mean confusion matrices of VGG11, DenseNet121, ResNet18, MobileNetV2, and EfficientNetB0 across five-fold cross-validation. For each architecture, the left panel shows mean classification counts and the right panel shows row-normalized proportions. Cell annotations represent the mean value $\pm$ standard deviation (SD) across the five folds. Rows correspond to the true class and columns to the predicted class.

3.3. Mixture-of-Experts Framework Performance

The MoE framework was evaluated using DenseNet121, MobileNetV2, and VGG11 as CNN backbones under the same stratified five-fold CV protocol. The validation performance of the four constituent experts and the final fusion outputs is summarized in **Table 7**. Across all three MoE configurations, the individual experts demonstrated consistent performance across folds, although clear differences were observed among expert contributions. Expert 3 achieved the highest validation accuracy across all backbone architectures, ranging from 0.950 to 0.954, indicating a strong specialization capability for its corresponding diagnostic category. Expert 4 showed the second-highest performance, with accuracies ranging from 0.861 to 0.866, whereas Experts 1 and 2 exhibited comparatively lower but stable performance, ranging from approximately 0.781 to 0.793.

At the fusion level, the VGG11-based MoE achieved the highest overall classification accuracy (0.676 ± 0.021), followed by the DenseNet121-based MoE (0.671 ± 0.033) and MobileNetV2-based MoE (0.662 ± 0.041). The fusion outputs also demonstrated moderate agreement with the reference labels, with Cohen's κ values of 0.541 ± 0.028 , 0.538 ± 0.037 , and 0.523 ± 0.046 for VGG11-MoE, DenseNet121-MoE, and MobileNetV2-MoE, respectively. Similarly, MCC values followed the same trend, reaching 0.546 ± 0.026 for VGG11-MoE, 0.543 ± 0.038 for DenseNet121-MoE, and 0.527 ± 0.048 for MobileNetV2-MoE. These results indicate that although individual experts showed different levels of specialization, the fusion network successfully integrated expert outputs to produce an integrated multiclass prediction.

Table 7. Validation performance of individual experts and final fusion outputs in the Mixture-of-Experts (MoE) frameworks across five-fold cross-validation. Values are reported as mean $\pm$ standard deviation (SD) across five folds. The highest value within each column is shown in bold.

MoE backbone	Expert 1 Accuracy	Expert 2 Accuracy	Expert 3 Accuracy	Expert 4 Accuracy	Fusion Accuracy	Cohen's κ	MCC
DenseNet121-MoE	0.791 $\pm$ 0.011	0.793 $\pm$ 0.026	0.951 $\pm$ 0.010	0.861 $\pm$ 0.006	0.671 $\pm$ 0.033	0.538 $\pm$ 0.037	0.543 $\pm$ 0.038
MobileNetV2-MoE	0.781 $\pm$ 0.023	0.783 $\pm$ 0.030	0.954 $\pm$ 0.015	0.864 $\pm$ 0.005	0.662 $\pm$ 0.041	0.523 $\pm$ 0.046	0.527 $\pm$ 0.048
VGG11-MoE	0.786 $\pm$ 0.026	0.786 $\pm$ 0.016	0.950 $\pm$ 0.013	0.866 $\pm$ 0.012	0.676 $\pm$ 0.021	0.541 $\pm$ 0.028	0.546 $\pm$ 0.026

*Abbreviations: κ , Cohen's kappa coefficient; MCC, Matthews correlation coefficient; MoE, Mixture-of-Experts; SD, standard deviation.

The class-wise performance of the three MOE configurations is summarized in **Table 8**. Across all three backbones, thyroiditis consistently showed the strongest classification performance, with F1-scores ranging from 0.845 to 0.855, whereas the normal class remained the most challenging, with F1-scores between 0.352 and 0.382. DenseNet121-MoE achieved the highest weighted F1-score (0.672 ± 0.024), while VGG11-MoE showed the highest macro-precision (0.677 ± 0.027) and weighted recall (0.676 ± 0.017). MobileNetV2-MoE yielded the highest thyroiditis precision (0.884 ± 0.029) and F1-score (0.855 ± 0.036). Overall, the three MoE

variants demonstrated similar performance profiles, with the largest differences occurring in the diffuse-goiter and normal classes.

Table 8. Class-wise classification performance of the DenseNet121-, VGG11-, and MobileNetV2-based Mixture-of-Experts (MoE) model across five-fold cross-validation. Precision, recall, and F1-score are reported as mean $\pm$ standard deviation (SD) across the five folds for each diagnostic class, together with macro- and weighted-average performance.

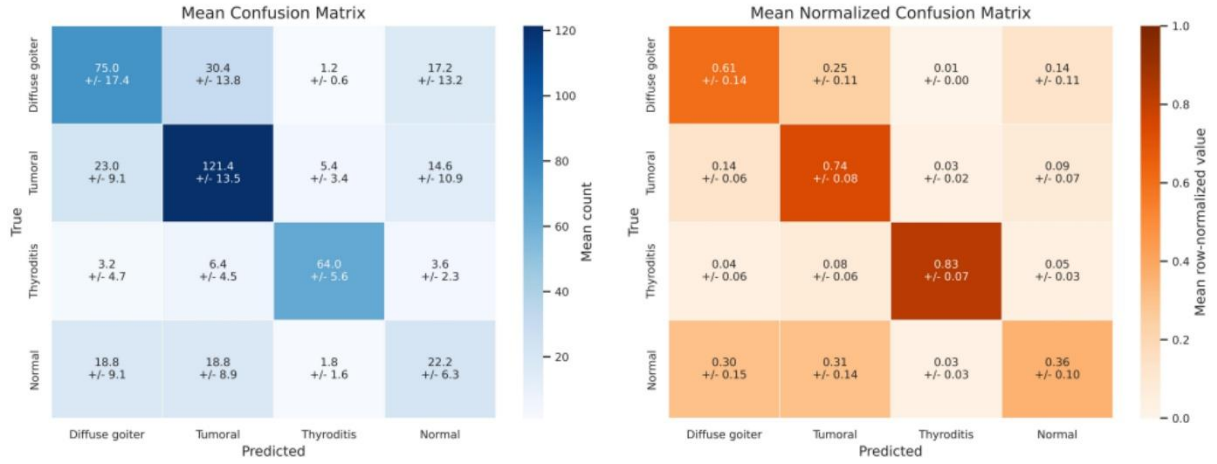
Model	Class / Average	Precision	Recall	F1-score
DenseNet121-MoE	Diffuse goiter	0.628 $\pm$ 0.053	0.662 $\pm$ 0.115	0.637 $\pm$ 0.034
	Tumoral	0.733 $\pm$ 0.071	0.714 $\pm$ 0.030	0.722 $\pm$ 0.037
	Thyroiditis	0.879 $\pm$ 0.023	0.826 $\pm$ 0.039	0.851 $\pm$ 0.023
	Normal	0.424 $\pm$ 0.102	0.377 $\pm$ 0.096	0.382 $\pm$ 0.027
	Macro average	0.666 $\pm$ 0.025	0.645 $\pm$ 0.025	0.648 $\pm$ 0.022
	Weighted average	0.685 $\pm$ 0.031	0.671 $\pm$ 0.027	0.672 $\pm$ 0.024
VGG11-MoE	Diffuse goiter	0.596 $\pm$ 0.040	0.711 $\pm$ 0.081	0.645 $\pm$ 0.030
	Tumoral	0.720 $\pm$ 0.044	0.720 $\pm$ 0.037	0.719 $\pm$ 0.029
	Thyroiditis	0.859 $\pm$ 0.051	0.834 $\pm$ 0.038	0.845 $\pm$ 0.031
	Normal	0.534 $\pm$ 0.174	0.292 $\pm$ 0.101	0.352 $\pm$ 0.070
	Macro average	0.677 $\pm$ 0.027	0.639 $\pm$ 0.027	0.640 $\pm$ 0.031
	Weighted average	0.682 $\pm$ 0.014	0.676 $\pm$ 0.017	0.667 $\pm$ 0.026
MobileNetV2-MoE	Diffuse goiter	0.632 $\pm$ 0.059	0.606 $\pm$ 0.112	0.612 $\pm$ 0.054
	Tumoral	0.692 $\pm$ 0.062	0.739 $\pm$ 0.067	0.711 $\pm$ 0.032
	Thyroiditis	0.884 $\pm$ 0.029	0.829 $\pm$ 0.057	0.855 $\pm$ 0.036
	Normal	0.410 $\pm$ 0.112	0.361 $\pm$ 0.085	0.373 $\pm$ 0.057
	Macro average	0.655 $\pm$ 0.042	0.633 $\pm$ 0.032	0.638 $\pm$ 0.032
	Weighted average	0.669 $\pm$ 0.038	0.662 $\pm$ 0.033	0.660 $\pm$ 0.033

The class-wise prediction behavior of the three MoE configurations is illustrated using mean confusion matrices across the five cross-validation folds (**Figure 5**). All three MoE models demonstrated a consistent classification pattern, with thyroiditis representing the most accurately identified category. The normalized confusion matrices showed high recall values for thyroiditis across MobileNetV2-MoE (0.83), DenseNet121-MoE (0.83), and VGG11-MoE (0.83). Tumoral lesions also achieved relatively strong recognition, with recall values of 0.74 for MobileNetV2-MoE and 0.71 for both DenseNet121-MoE and VGG11-MoE.

Diffuse goiter classification showed moderate performance, with normalized recalls of 0.61, 0.66, and 0.66 for MobileNetV2-MoE, DenseNet121-MoE, and VGG11-MoE, respectively. In contrast, normal scintigraphy remained the most difficult category, although the MoE models showed improved recognition compared with the corresponding baseline patterns, achieving normalized recalls of 0.36 for MobileNetV2-MoE and 0.38 for both DenseNet121-MoE and VGG11-MoE. The main source of error for normal cases was confusion with diffuse goiter and tumoral classes, suggesting overlapping scintigraphic characteristics between normal and abnormal uptake patterns.

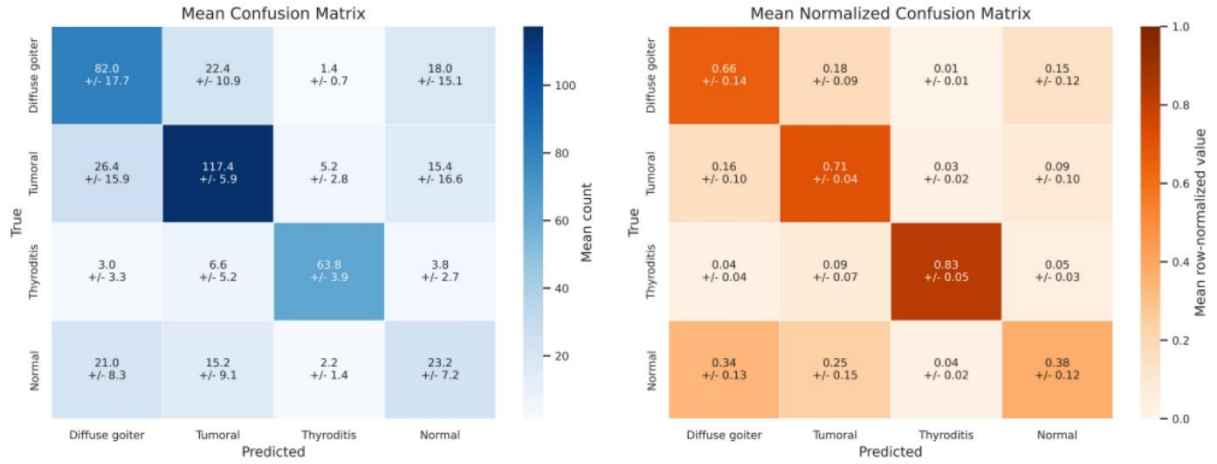
MobileNetV2-MoE

MoE Confusion Matrix: Mean $\pm$ 95% CI Across 5 Folds



DenseNet121-MoE

MoE Confusion Matrix: Mean $\pm$ 95% CI Across 5 Folds



VGG11-MoE

MoE Confusion Matrix: Mean $\pm$ 95% CI Across 5 Folds

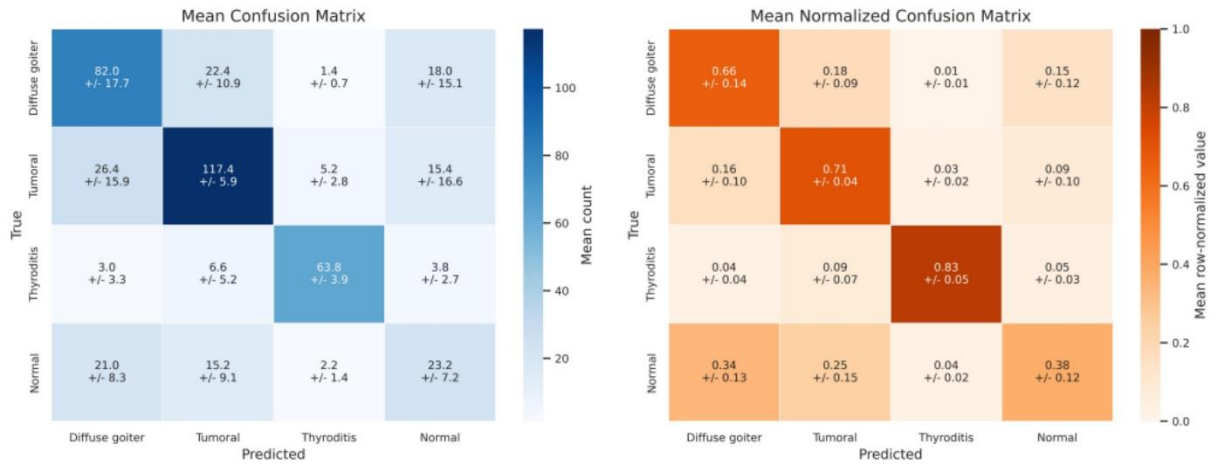


Figure 5. Mean confusion matrices of the DenseNet121-, MobileNetV2-, and VGG11-based Mixture-of-Experts (MoE) models across five-fold cross-validation. The left panels show mean classification counts, whereas the right panels show row-normalized confusion matrices. Values represent the mean $\pm$ SD across folds.

Thyroiditis showed the highest classification consistency across all MoE configurations, whereas normal scans remained the most challenging category.

To investigate the contribution of individual experts within the MoE framework, the distributions of expert logits for correctly classified samples were analyzed (**Figure 6**). The logit distributions demonstrated clear expert specialization patterns across diagnostic categories. Expert 3 showed dominant positive activation for thyroiditis cases across all backbone configurations, indicating its strong contribution to thyroiditis discrimination. Similarly, Expert 2 demonstrated strong responses for tumoral samples and contributed substantially to tumoral classification.

For diffuse goiter and normal categories, the separation among experts was less pronounced, reflecting the lower class-wise discrimination observed in the confusion matrices. Nevertheless, the combination of complementary expert responses enabled the fusion model to integrate multiple decision patterns rather than relying on a single feature representation.

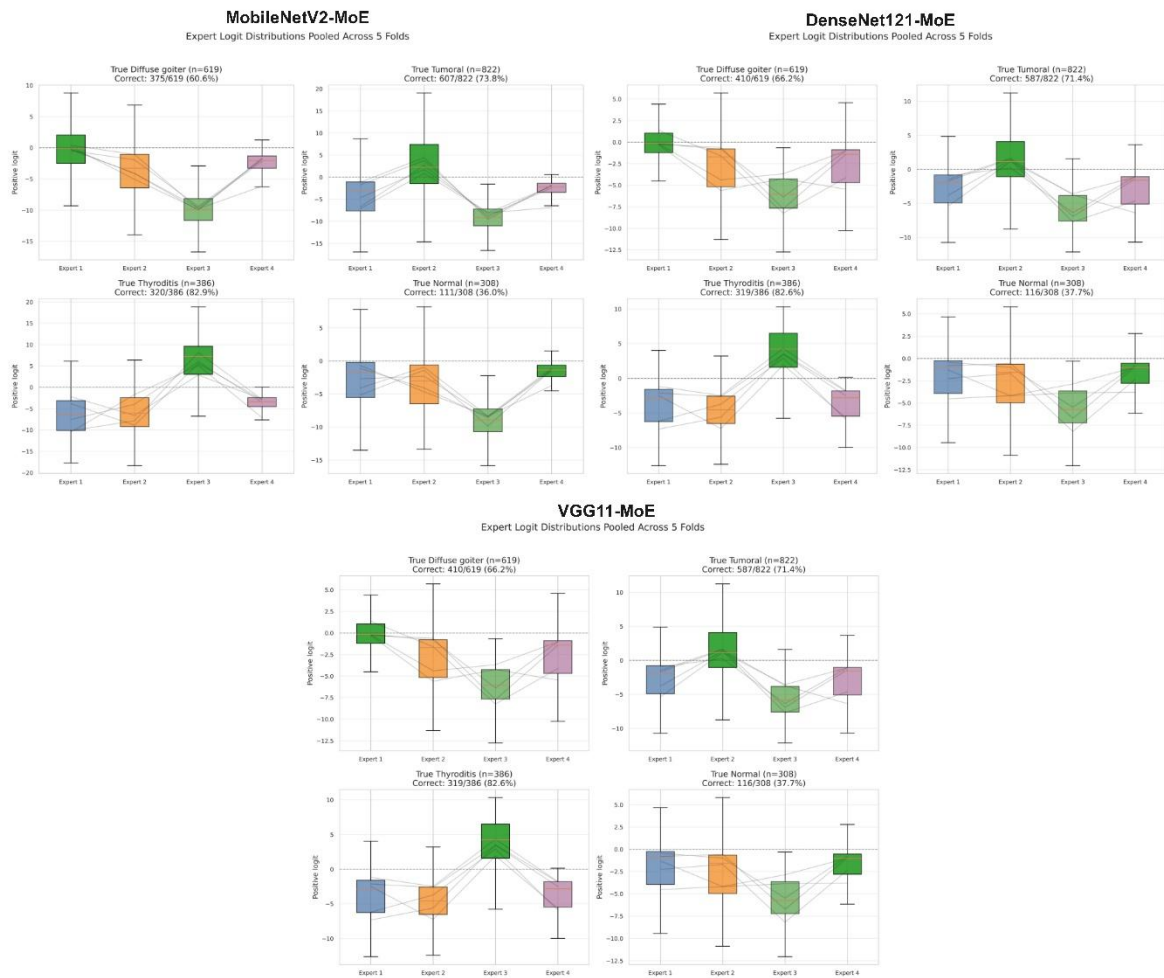


Figure 6. Expert specialization patterns in the Mixture-of-Experts framework based on mean positive-class logits. Rows represent the true diagnostic classes, and columns correspond to the four class-specific experts: Expert 1 (diffuse goiter), Expert 2 (tumoral disease), Expert 3 (thyroiditis), and Expert 4 (normal scintigraphy). Cell values indicate the mean positive-class logit $\pm$ SD across five-fold cross-validation. Higher logit values indicate stronger expert activation toward the corresponding class. The heatmaps demonstrate the learned

specialization and complementary contribution of individual experts, while the final classification decision was obtained by the fusion network using the joint expert-logit representation.

The average expert logit maps across diagnostic classes are summarized in **Figure 7**. These heatmaps provide an overview of the learned specialization of each expert. Across all three MoE architectures, Expert 3 exhibited the strongest positive activation for thyroiditis, whereas Expert 2 showed increased responses for tumoral cases. The remaining experts demonstrated complementary but weaker activation patterns across classes.

The consistency of these patterns across DenseNet121-, MobileNetV2-, and VGG11-based MoE models suggests that the fusion mechanism successfully preserved expert diversity and combined heterogeneous decision signals. This complementary behavior provides an explanation for the final MoE predictions beyond the output of an individual CNN classifier.

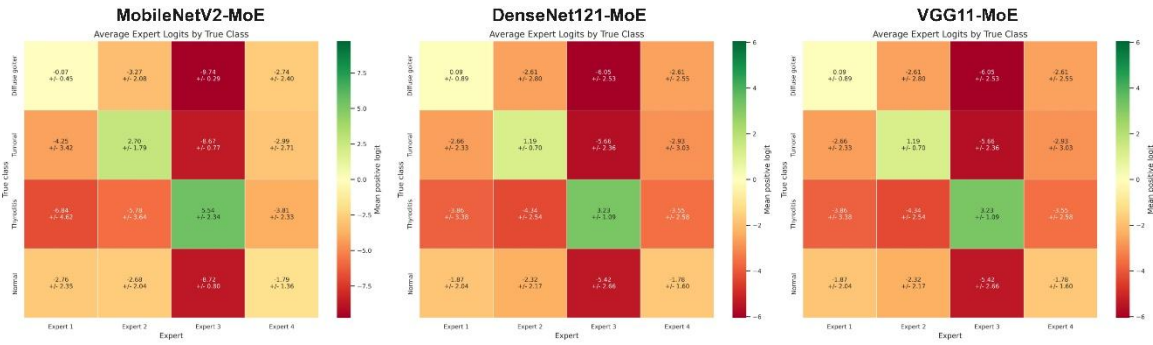


Figure 7. Average expert logits by true diagnostic class for the three MoE architectures. Heatmaps represent mean expert responses across five-fold cross-validation, illustrating the specialization and complementary contribution of individual experts within the fusion framework.

3.4. Explainability Analysis Using Grad-CAM

In order to gain insight into the decision-making process in the case of the baseline CNN, the Grad-CAM visualization technique was applied to a representative sample of correctly and incorrectly classified examples using the DenseNet121 model. This analysis aimed to identify the image parts that contribute most to the model's decision-making process and see whether the attention patterns learned by the model corresponded to the diagnostically important thyroid uptake regions. **Figure 8** represents examples of correctly and incorrectly classified cases for the baseline experiment.

As can be seen from **Figure 8**, the Grad-CAM heat maps tend to focus on the regions related to the thyroid uptake rather than on the irrelevant background regions. This means that the CNN learned the image features associated with the patterns of thyroid scintigraphy. In the case of correctly classified diffuse goiter and normal examples, attention is focused around the thyroid, while misclassified examples have the attention distributed more broadly or shifted from the thyroid region.

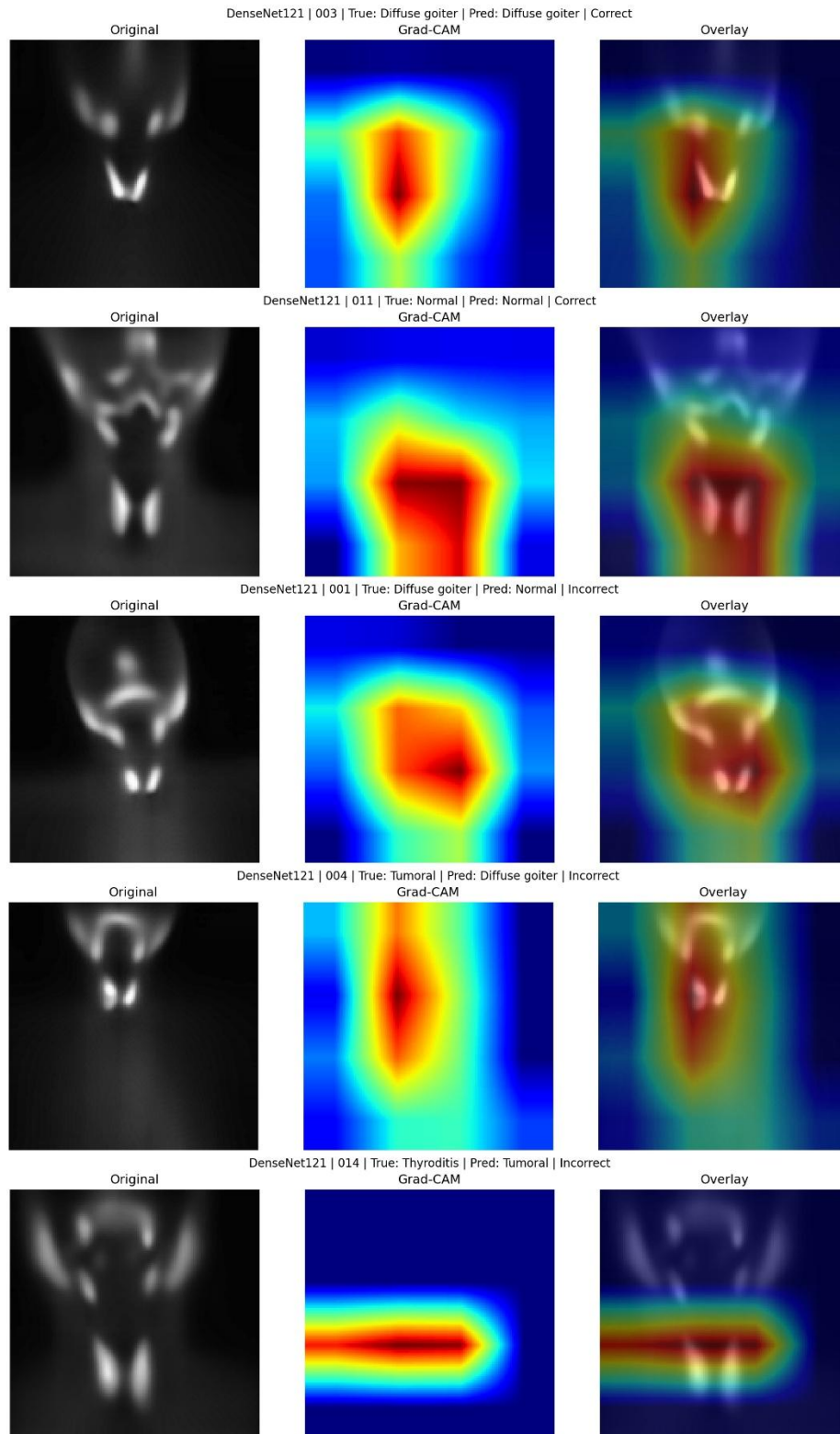


Figure 8. Grad-CAM output for the DenseNet121 classification for representative images from the baseline test run. Each row displays the input thyroid scintigraphy image, Grad-CAM heatmap, and the overlay of the heatmap on the input image. The correctly classified images from the diffuse goiter class and normal class, as well as the incorrectly classified images, are shown to demonstrate the attention focus of the model.

The visual examination of the MoE decision-making process was carried out by using the Grad-CAM visualization technique on the individual expert networks. **Figures 9 and 10** demonstrate typical attention maps obtained from different experts belonging to classes and making either correct or wrong decisions. Compared with the previous experiments, the MoE maps qualitatively showed more structured, class-specific activation patterns, resulting from the separate learning of each expert. The diffuse goiter, tumoral, thyroiditis, and normal detectors were mostly concerned with the thyroid uptake zones related to their categories, implying that the process of learning made the model's decisions more understandable.

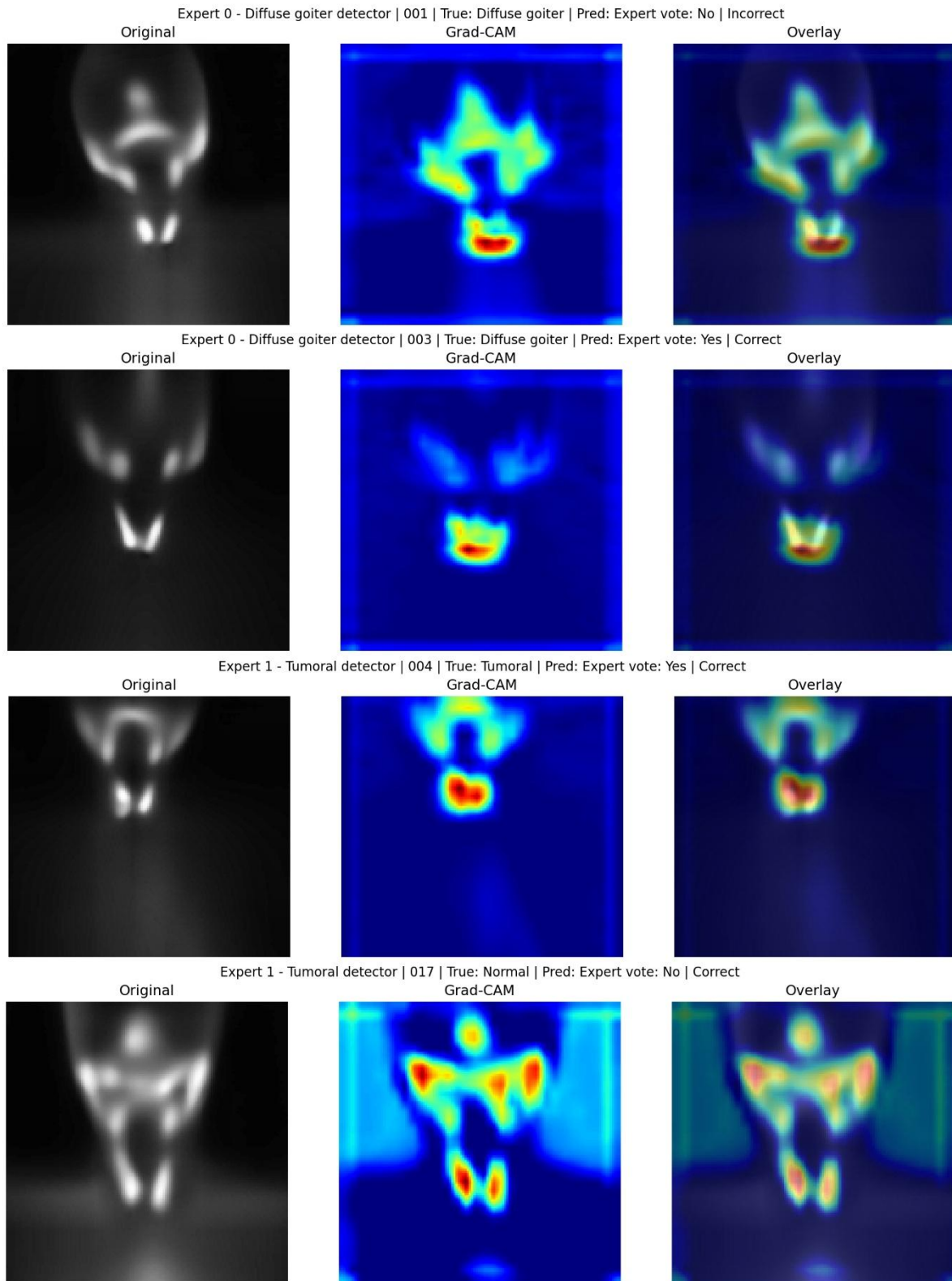


Figure 9. Grad-CAM representation of the MoE architecture for thyroiditis and normal class-specific experts. In each row, the original scintigraphy image (left), the Grad-CAM heatmap (middle), and the heatmap overlay (right) are shown. The representations reflect the attention patterns of the experts for correctly and wrongly classified cases, illustrating the discriminative regions learned by each expert.

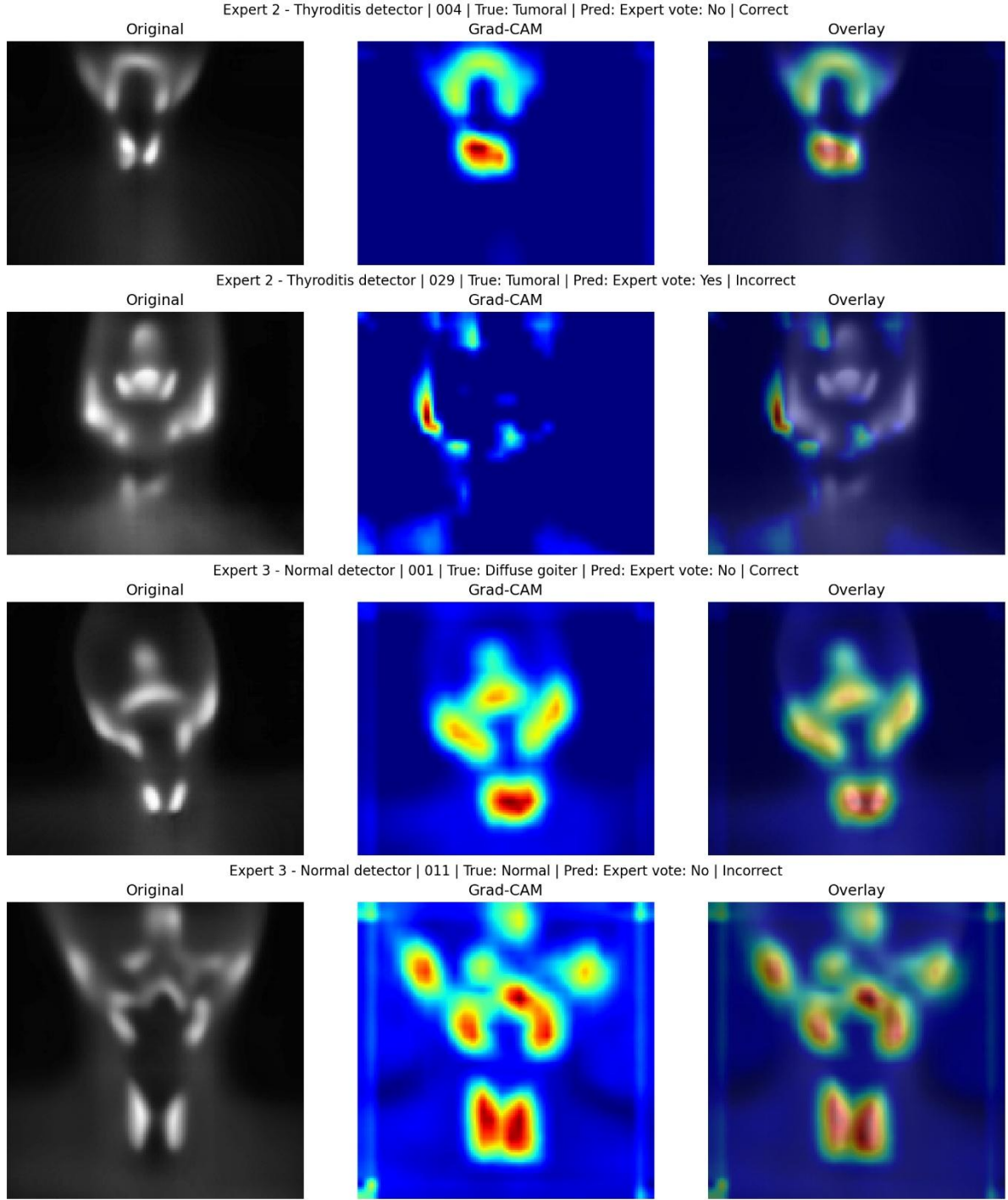


Figure 10. Grad-CAM representation of the MoE architecture for diffuse goiter and tumoral class-specific experts. In each row, the original scintigraphy image (left), the Grad-CAM heatmap (middle), and the heatmap overlay (right) are shown. The attention patterns of the experts illustrate their localization to thyroid uptake regions, giving an idea of their specialized functioning.

3.5. Effect of Classification Granularity on Model Performance

In addition to the four-class classification task, a secondary binary classification analysis was performed by regrouping the diagnostic categories into tumoral and non-tumoral classes. This analysis was conducted to evaluate the ability of the developed MoE model to discriminate tumoral-related scintigraphic patterns from

non-tumoral conditions and to determine the impact of reducing the number of diagnostic categories on model performance. The binary classification results across the five-fold CV experiments are summarized in **Table 9**.

The binary classification analysis demonstrated improved discrimination between tumoral and non-tumoral categories compared with the more challenging four-class classification setting. DenseNet121 achieved the highest overall discriminative performance, with the highest mean AUC (0.843 ± 0.029), sensitivity (0.683 ± 0.050), balanced accuracy (0.776 ± 0.022), and tumoral F1-score (0.722 ± 0.030). VGG11 achieved a comparable overall accuracy (0.798 ± 0.010) with the highest specificity (0.902 ± 0.024), indicating strong recognition of non-tumoral cases. Across all architectures, specificity remained higher than sensitivity, suggesting that the models were more consistent in excluding non-tumoral cases than detecting all tumoral cases.

Table 9. Binary classification performance for tumoral versus non-tumoral discrimination across five-fold cross-validation. Values are presented as mean $\pm$ standard deviation (SD) across five-fold cross-validation.

Model	Accuracy	AUC	Sensitivity (Recall)	Specificity or TNR	Precision or PPV (Tumoral)	NPV (Tumoral)	F1-score (Tumoral)	Balanced Accuracy	MCC
VGG11	0.798 ± 0.010	0.821 ± 0.017	0.633 ± 0.042	0.902 ± 0.024	0.804 ± 0.034	0.797 ± 0.016	0.707 ± 0.020	0.767 ± 0.013	0.566 ± 0.021
DenseNet121	0.798 ± 0.019	0.843 ± 0.029	0.683 ± 0.050	0.870 ± 0.031	0.768 ± 0.037	0.815 ± 0.022	0.722 ± 0.030	0.776 ± 0.022	0.567 ± 0.041
ResNet18	0.781 ± 0.025	0.819 ± 0.025	0.611 ± 0.067	0.888 ± 0.028	0.775 ± 0.041	0.786 ± 0.028	0.681 ± 0.047	0.749 ± 0.031	0.529 ± 0.057
MobileNetV2	0.795 ± 0.020	0.832 ± 0.014	0.633 ± 0.054	0.896 ± 0.011	0.792 ± 0.017	0.796 ± 0.023	0.703 ± 0.038	0.765 ± 0.025	0.558 ± 0.043
EfficientNetB0	0.789 ± 0.018	0.834 ± 0.027	0.644 ± 0.065	0.880 ± 0.032	0.774 ± 0.039	0.799 ± 0.026	0.701 ± 0.035	0.762 ± 0.024	0.548 ± 0.041

Abbreviations: AUC, area under the curve; **F1-score**, F1 measure; **MCC**, Matthews correlation coefficient; **NPV**, negative predictive value; **PPV**, positive predictive value; **TNR**, true negative rate.

The performance of the DenseNet121 model (as the best model) for binary discrimination between tumoral and non-tumoral thyroid scintigraphy cases is illustrated in **Figure 11**. The confusion matrix demonstrated that the

model correctly classified the majority of cases in both categories, with a mean of 228.4 ± 10.1 correctly identified non-tumoral cases and 112.2 ± 9.9 correctly identified tumoral cases across five-fold CV. Misclassification occurred mainly in the tumoral category, with 52.2 ± 10.4 tumoral cases incorrectly classified as non-tumoral, whereas only 34.2 ± 10.1 non-tumoral cases were incorrectly predicted as tumoral.

The ROC analysis further confirmed the discriminative capability of DenseNet121 for separating tumoral from non-tumoral cases. The model achieved an AUC of 0.843 for both non-tumoral and tumoral classes, indicating good classification ability after reducing the original four-class problem into a binary decision task.

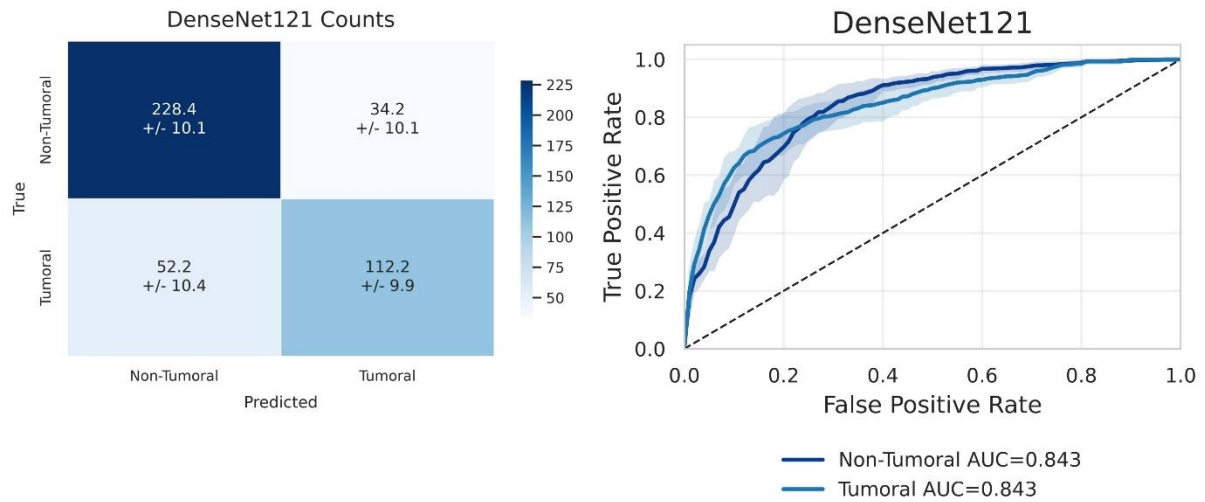


Figure 11. Binary classification performance of the DenseNet121 model for tumoral versus non-tumoral thyroid scintigraphy discrimination. The left panel presents the mean confusion matrix across five-fold cross-validation, with values representing mean classification counts $\pm$ standard deviation. The right panel shows the receiver operating characteristic (ROC) curves for the two classes, demonstrating the model’s discriminative performance with an AUC of 0.843 for both tumoral and non-tumoral categories.

DISCUSSION

Thyroid scintigraphy provides unique functional information about thyroid uptake patterns; however, automated interpretation remains challenging due to overlapping disease characteristics, multicenter acquisition variability, differences in imaging protocols, limited labeled data, and class imbalance. Therefore, robust AI systems require evaluation under realistic clinical conditions beyond performance metrics alone. To the best of our knowledge, this is the first study applying a class-specific MoE strategy to thyroid scintigraphy analysis, providing a new framework for interpretable multiclass classification. This study investigated DL-based thyroid scintigraphy classification in a challenging multicenter setting by comparing conventional CNN models with a class-specific MoE framework. Unlike standard end-to-end classifiers that learn a shared representation for all categories, the proposed MoE approach uses specialized experts for individual diagnostic classes and integrates their outputs through a fusion network.

The main contributions of this study include: (1) evaluation of a multicenter thyroid scintigraphy dataset with real-world variability across four diagnostic categories; (2) systematic comparison of CNN and MoE models under identical five-fold CV settings; (3) application of GAN-based augmentation to address class imbalance

while restricting synthetic data to the training process; (4) implementation of strict leakage prevention through fixed folds, fold-specific preprocessing, and training-only augmentation; and (5) assessment of a simplified tumoral versus non-tumoral classification task to investigate the effect of diagnostic complexity on model performance. Beyond predictive performance, this study also investigated model interpretability through XAI analysis and expert-level response evaluation, providing insight into how individual experts contribute to the final decision. Furthermore, the dataset, experimental configuration, and source code are publicly available to support reproducibility and future benchmarking of AI methods for thyroid scintigraphy analysis.

The MoE framework achieved comparable multiclass performance to conventional CNN models, with the VGG11-based MoE achieving the highest fusion accuracy (0.676 ± 0.021), while DenseNet121 provided the highest baseline CNN accuracy (0.674 ± 0.023). These results indicate that increased architectural specialization does not necessarily lead to substantial gains in overall accuracy for complex multiclass clinical classification. However, the MoE framework provided additional value through expert-level analysis, revealing different levels of specialization among diagnostic categories and highlighting variations in classification difficulty. Therefore, the benefit of MoE extends beyond performance improvement alone by providing a more structured and interpretable decision process. In medical AI, particularly nuclear medicine, transparency and understanding of model behavior are essential for clinical acceptance [32-34]. The combination of expert-response analysis and Grad-CAM visualization demonstrated that model decisions were primarily driven by clinically relevant thyroid uptake regions, supporting the plausibility of the learned representations.

Beyond quantitative evaluation, visualization of individual cases provides an additional perspective on whether DL models rely on clinically meaningful scintigraphic patterns. Figure 8 presents representative case-level interpretability results for the baseline CNN models using Grad-CAM visualization. These examples demonstrate the regions of the input images that contributed most strongly to the model predictions. In correctly classified cases, the activation maps were mainly localized within the thyroid gland, suggesting that the models learned relevant functional imaging characteristics associated with tracer distribution and uptake patterns rather than relying predominantly on irrelevant background information. However, the interpretation of conventional CNN models remains limited by their shared representation strategy. Since a single CNN feature extractor is optimized simultaneously for discrimination among all diagnostic categories, the generated activation maps represent an aggregated decision process. Although the highlighted regions may correspond to diagnostically meaningful areas, these visualizations do not directly indicate which disease-specific characteristics contributed to the final classification. This limitation is particularly important in thyroid scintigraphy, where several diagnostic entities may demonstrate partially overlapping uptake patterns.

Figures 9–10 demonstrate the corresponding case-level analysis for the proposed MoE framework. Unlike conventional CNN models, the MoE architecture separates the classification process into multiple class-specific experts, where each expert is trained to recognize one diagnostic category against the remaining classes. Therefore, the interpretability analysis provides not only spatial information through Grad-CAM but also insight into the response of individual diagnostic pathways before the final fusion step.

The MoE visualizations show that different experts generated distinct activation patterns depending on the diagnostic characteristics of the presented case. For example, the diffuse goiter and tumoral experts

demonstrated stronger responses in regions associated with altered thyroid uptake patterns relevant to their corresponding diagnostic categories. In cases where the final prediction was correct, the dominant expert response was generally consistent with the assigned diagnosis, suggesting that the fusion network benefited from combining specialized expert outputs rather than relying on a single shared representation.

From a clinical perspective, this expert-based interpretation provides an additional advantage over conventional CNN models. A nuclear medicine physician does not evaluate scintigraphy images by considering a single generalized abnormality pattern; rather, the interpretation involves evaluating multiple possible diagnostic explanations and comparing their supporting imaging characteristics. The MoE framework follows a similar principle by decomposing the decision process into multiple diagnostic-specific responses, allowing examination of which expert contributed most strongly to the final prediction.

Nevertheless, the difference between CNN and MoE explanations should not be interpreted as one framework producing universally superior Grad-CAM maps. Grad-CAM highlights regions associated with model predictions but does not completely describe the underlying reasoning process. Instead, the main advantage of the MoE framework is the additional structural interpretability provided by expert-specific predictions and logits. By combining expert-level responses with spatial explanations, the proposed approach provides a more transparent view of model decision formation.

Overall, the qualitative findings from Figures 8–11 complement the quantitative results by demonstrating that both baseline CNN and MoE models learned clinically relevant scintigraphic features. However, the MoE framework extends interpretability beyond localization alone by providing a class-specific diagnostic decomposition, which may facilitate greater transparency and confidence in future AI-assisted nuclear medicine applications.

Qiao et al. [11] investigated 1,430 thyroid scintigraphy examinations acquired at a single institution and from a single imaging system. Their original dataset consisted of only three diagnostic classes: 175 normal subjects, 834 patients with Graves' disease, and 421 patients with subacute thyroiditis. The study subsequently used a progressively growing auxiliary-classifier GAN to generate synthetic images until each class contained 1,000 samples. In addition, ImageNet-pretrained networks were used for transfer learning, with previous layers frozen and only the final classification component fine-tuned. The best ResNet model consequently produced class-specific accuracies of 91.78%, 92.78%, and 89.00% for normality, Graves' disease, and subacute thyroiditis, respectively. These results are impressive, but the classification problem was fundamentally different from the current one. Only three relatively distinct categories were considered, and the data originated from one hospital and one scanner. Each of these conditions may reduce the effective difficulty of model training compared with learning four heterogeneous classes from scratch on pooled multicenter data.

Ma et al. [35] likewise reported strong results using a single-center SPECT dataset. Their dataset contained 780 Graves' disease, 438 Hashimoto disease, 810 subacute thyroiditis, and 860 normal examinations, corresponding to 2,888 images in four classes. All images were obtained from a single hospital using a Siemens SPECT ECAM system. Moreover, their training strategy combined data augmentation, transfer learning, a modified DenseNet architecture, and an optimization-based learning-rate procedure. Thus, although their sample size was comparable to or somewhat larger than that of the current work, the acquisition environment was substantially

more homogeneous. The relatively balanced disease distribution, single-center imaging source, and different disease categories provide plausible explanations for why high classification performance could be achieved without necessarily implying that the same performance would be maintained under the heterogeneous five-center setting considered here.

Yang et al. [12] provide a particularly useful comparison because their study addressed four-class classification and included external validation. They used 3,087 images from one center for model development, divided into 2,468 training and 619 internal validation cases, together with 302 examinations from a second center for external testing. The internal training set comprised 1,040 diffusely increased, 902 diffusely decreased, 97 focally increased, and 429 heterogeneous-uptake cases, whereas the external dataset contained 120, 49, 13, and 120 cases, respectively. InceptionV3 achieved an overall accuracy of 92.73% internally and 87.75% externally, with class-specific external AUC values ranging from 0.915 to 1.000. However, Yang et al. [12] classified uptake patterns as diffusely increased, diffusely decreased, focally increased, and heterogeneous uptake, rather than the heterogeneous pathology-based class combinations used in the current study. These differences are important because uptake-pattern classification provides visually defined targets that may have stronger direct correspondence with image appearance than diagnostic groups containing several different scintigraphy phenotypes. The reduction in performance from 92.73% internally to 87.75% at the second center also demonstrates that even a strong model experiences a measurable performance decrease when acquisition conditions change.

Zhao et al. [36] reported some of the highest results among the attached thyroid scintigraphy studies. Their dataset consisted of 3,194 images obtained from three hospitals, with 2,067 images for training, 514 for internal validation, and 613 for external validation. Importantly, the four disease groups were almost perfectly balanced: 808 Graves' disease cases, 742 normal cases, 826 subacute thyroiditis cases, and 818 thyroid tumors. Their modified ResNet34 achieved an internal accuracy of 0.944 and an AUC of 0.992, while external accuracy remained 0.931. Several characteristics may account for these high values. First, the near-balanced class distribution provides a considerably more favorable optimization setting than the 341/466/924/717 distribution in the current study. Second, poor-quality examinations were excluded, and images were reviewed by experienced nuclear medicine physicians before inclusion. Third, their classes- Graves' disease, normality, subacute thyroiditis, and thyroid tumor- were defined as four direct diagnostic groups instead of consolidating multiple original diagnoses into heterogeneous classes. Therefore, although Zhao et al. [36] provide convincing evidence that high accuracy is achievable in multicenter thyroid scintigraphy, their reported accuracy should not be considered directly equivalent to performance on the present benchmark.

A useful counterpoint is provided by the Thyroidiomics study of Sabouri et al. [13] which explicitly tested center generalizability. Their dataset contained 2,643 thyroid scintigraphy images collected from nine centers and categorized into three pathology classes: multinodular goiter, thyroiditis, and diffuse goiter. Rather than randomly pooling centers, the authors used leave-one-center-out CV, training on eight centers and evaluating on the unseen ninth center. Under this considerably more demanding evaluation, physician-ROI-based classification achieved an accuracy of approximately 0.76 and ROC-AUC of approximately 0.92, while the fully automated segmentation-based approach achieved approximately 0.74 accuracy and 0.90 ROC-AUC. This study is especially informative for interpreting the current findings because its accuracy is much closer to the range

observed here despite using only three classes. It demonstrates that substantial heterogeneity and genuine center-wise generalization can markedly reduce accuracy compared with results obtained from more controlled or randomly pooled datasets.

Interestingly, the class-specific pattern reported by Sabouri et al. [13] is also consistent with the current observations. Thyroiditis was their strongest classification class, and thyroiditis was also the most distinguishable class throughout the CNN and MoE experiments in the present work. This convergence across different modeling strategies suggests that the characteristic reduction or alteration in thyroid tracer uptake associated with thyroiditis may provide stronger imaging separation than classes involving nodular or morphologically heterogeneous patterns. Conversely, performance for more heterogeneous categories can be considerably lower even when overall AUC remains high. This reinforces the importance of reporting macro and class-specific metrics rather than judging performance through accuracy alone.

Taken together, comparison with previous work suggests that the numerical difference between the current results and published high-accuracy models cannot reasonably be attributed to architecture alone. Dataset composition is at least equally important. Previous studies demonstrating accuracies approaching or exceeding 90% commonly benefited from one or more favorable characteristics: single-center acquisition, predefined and visually distinct diagnostic or uptake-pattern classes, nearly balanced datasets, and exclusion of difficult or poor-quality cases.

The binary classification experiment offered further understanding about the impact of diagnostic granularity on the performance of the model. When combining four diagnostic types into two groups, tumoral and non-tumoral, the DenseNet121 baseline demonstrated improved discriminatory ability, with the ROC-AUC value equal to 0.843, as opposed to the multi-class classification scenario. Nevertheless, the challenge of detecting tumoral cases indicates that some scintigraphic patterns possess analogous functional properties for various thyroid diseases. Consequently, while it is easier for the model to learn in the case of reduced classes, it does not address the inherent uncertainty of scintigraphy interpretation for thyroid.

Beyond diagnostic classification, the proposed MoE framework may have broader relevance for future theranostic applications, where interpretable AI is essential for clinical translation. Recent studies have highlighted that although AI can support multiple stages of theranostic workflows, including image analysis, segmentation, and personalized treatment planning, limited explainability remains a major challenge [37, 38]. By incorporating class-specific experts and visualization of expert decision patterns, MoE architectures may provide a more transparent alternative to conventional black-box models, potentially supporting future AI-assisted applications in molecular imaging and radiopharmaceutical therapy.

Despite these contributions, several limitations should be acknowledged. First, although the dataset included multiple centers, external validation on completely independent institutions was not performed and remains necessary before clinical deployment. Second, GAN-generated images were used only for training augmentation, and although diversity monitoring was applied, synthetic data cannot fully replace the acquisition of additional real clinical cases. Third, the current framework relies exclusively on planar scintigraphy images and does not incorporate additional clinical information such as laboratory findings, ultrasound characteristics, or patient demographics, which may further improve diagnostic performance.

Overall, this study demonstrates that an explainable class-specific MoE framework combined with GAN-based augmentation provides a structured approach for multiclass thyroid scintigraphy analysis in a heterogeneous multicenter environment. While MoE did not substantially exceed conventional CNNs in overall accuracy, it provided additional interpretability through expert specialization analysis and offered a framework better aligned with the complexity of clinical diagnostic reasoning. The current models should be considered as decision-support systems rather than replacements for specialist evaluation. The publicly available dataset, source code, and experimental configuration may facilitate future research and independent benchmarking of explainable AI methods in nuclear medicine imaging. Future studies involving independent external validation, prospective clinical testing, and integration with complementary clinical information are warranted to determine their role in routine nuclear medicine practice.

CONCLUSION

This study developed and evaluated an explainable DL framework for four-class classification of thyroid scintigraphy in a heterogeneous multicenter setting. By incorporating class-specific expert networks within a MoE architecture, the proposed approach provided a mechanism for modeling distinct diagnostic patterns rather than relying on a single unified classifier. The analysis demonstrated that individual experts learned meaningful specialization behaviors, and their integration through a fusion network produced an integrated multiclass prediction strategy. The findings also highlight the complexity of automated thyroid scintigraphy interpretation. Although DL models achieved promising overall discrimination, performance varied substantially among diagnostic categories, with thyroiditis consistently showing strong classification characteristics and normal scintigraphy remaining the most difficult category. These observations indicate that accuracy-based evaluation alone may not fully represent clinical applicability, particularly in settings where misclassification patterns differ across disease groups.

ABBREVIATIONS

AI: Artificial intelligence; **AUC:** Area under the receiver operating characteristic curve; **BN:** Batch normalization; **CAD:** Computer-aided diagnosis; **CNN:** Convolutional neural network; **CV:** Cross-validation; **DCGAN:** Deep convolutional generative adversarial network; **DL:** Deep learning; **DICOM:** Digital Imaging and Communications in Medicine; **GAN:** Generative adversarial network; **Grad-CAM:** Gradient-weighted Class Activation Mapping; **LEAP:** Low-energy all-purpose; **LEHR:** Low-energy high-resolution; **MCC:** Matthews correlation coefficient; **MLP:** Multilayer perceptron; **MoE:** Mixture-of-Experts; **NAFNet:** Nonlinear activation-free network; **ROC:** Receiver operating characteristic; **SD:** Standard deviation; **SPECT:** Single-photon emission computed tomography; **TSH:** Thyroid-stimulating hormone; **XAI:** Explainable artificial intelligence.

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Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Author contributions

EY: Conceptualization, supervision, project administration, methodology, data curation, formal analysis, validation, visualization, and writing the original draft.

MN: Methodology, formal analysis, validation, visualization, and writing the original draft preparation.

FE: Clinical expertise, clinical data validation, and clinical data interpretation.

NKZ: Clinical expertise, clinical data validation, and clinical data interpretation.

ZV: Data curation and clinical data collection.

SB: Data curation and clinical data collection.

HV: Supervision, data curation, and clinical data collection.

SRK: Conceptualization, supervision, methodology, and validation.

All authors reviewed and approved the final version of the manuscript.

Data availability

The de-identified thyroid scintigraphy dataset analyzed in this study, along with the source code required to reproduce the preprocessing pipeline, model development, training procedures, and main experimental analyses, will be publicly released upon publication of this article at: <https://elmirayazdani.github.io/thyroid-scintigraphy-xai/>. The repository will contain the thyroid scintigraphy images, corresponding diagnostic labels, documentation of the original four-class diagnostic scheme used for model development, and implementations of the baseline convolutional neural network (CNN) and mixture-of-experts (MoE) frameworks. In addition, relevant configuration files and instructions for reproducing the reported experiments will be provided to facilitate transparency and independent validation of the findings.

Ethical approval

This retrospective multicenter study was conducted in accordance with the ethical principles of the Declaration of Helsinki and applicable institutional requirements for research involving human data. The study protocol was reviewed and approved by the Ethics Committee of the Iran University of Medical Sciences (IUMS) under ethics approval IR.IUMS.REC.1405.163.

Consent to participate

The requirement for informed consent was waived by the Iran University of Medical Sciences (IUMS) Review Board due to the retrospective nature of the study.

Consent to publish

Not applicable. This study used retrospectively collected, de-identified thyroid scintigraphy images, and no identifiable individual data are presented. Therefore, consent for publication was not required.

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